



COMSATS University Islamabad, Sahiwal Campus

Itinera

**A project presented to
COMSATS University Islamabad Sahiwal Campus**

**In partial fulfillment
of the requirement for the degree of**

Bachelor of Science in Computer Science (2022-2026)

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Executive Summary

Itinera is a full-stack AI-powered travel planning and marketplace platform developed to simplify trip creation, travel management, and travel package booking through an intelligent digital environment. The system enables users to generate personalized travel itineraries by entering details such as origin, destination, budget, duration, and group size. Based on these inputs, the platform produces AI-generated travel plans including hotel suggestions, day-wise itineraries, activities, recommendations, locations, and coordinates.

The platform also provides smart travel support features such as live destination weather forecasts, weather-based travel alerts, packing suggestions, travel document checklists, and printable PDF travel reports. Users can further improve their generated trips through an AI trip re-optimizer, which allows them to modify plans according to specific needs such as reducing cost, adding rainy-day alternatives, minimizing walking, making the trip family friendly, or optimizing it for food experiences.

In addition to AI trip planning, Itinera includes a complete travel marketplace where agencies and collaborators can publish travel packages and paid travel plans. Users can browse marketplace listings, purchase plans through a dummy checkout flow, manage bookings, and submit reviews and ratings. The platform follows a commission-based business model, where the system earns a commission from each marketplace order.

Itinera supports multiple user roles including Normal User, Admin, Agency Partner, and Collaborator/Travel Blogger. Admins can manage users, approve partner applications, monitor marketplace activity, track revenue analytics, and manage orders. Partners and collaborators are provided with dedicated dashboards to publish packages, track earnings, and manage their travel offerings. The system is built using Next.js 15, React, TypeScript, Tailwind CSS, ShadCN UI, Convex, Clerk Authentication, OpenAI SDK, Open-Meteo API, Pexels API, Mapbox, Arcjet, and Resend, ensuring a scalable, secure, and modern AI-based travel platform.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DB	Database
FYP	Final Year Project
GUI	Graphical User Interface
HTTP	HyperText Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
PDF	Portable Document Format
REST	Representational State Transfer
UI	User Interface
UX	User Experience
SDK	Software Development Kit
RBAC	Role-Based Access Control
LLM	Large Language Model

Table of Contents

Chapter 1	13
INTRODUCTION.....	13
1 Introduction	1
1.1 Relevance to Course Module.....	2
1.2 Project Background	2
1.3 Literature Review	3
1.4 Analysis from Literature Review	4
1.4.1 Key Findings.....	4
1.4.2 Conceptualization of Itinera	4
1.5 Methodology and Software Lifecycle	5
1.5.1 Key Phases	5
1.5.2 Rationale behind Selected Methodology	6
Chapter 2	7
PROBLEM DEFINITION	7
2 Problem Definition	7
2.1 Problem Statement.....	7
2.2 Deliverables and Development Requirements.....	8
2.2.1 Major Deliverables:	8
2.2.2 Development Requirements	9
2.3 Current System	9
Chapter 3	11
REQUIREMENTS ANALYSIS.....	11
3 Requirement Analysis.....	11

3.1 Use Case Diagram	11
3.2 Use Case Description	13
3.3 Functional Requirements	24
3.3.1 User Login and Signup	24
3.3.2 AI Trip Generation	24
3.3.3 Trip Management	25
3.3.4 Weather and Travel Support	25
3.3.5 PDF Travel Report.....	25
3.3.6 AI Trip Re-Optimization.....	25
3.3.7 Marketplace Browsing.....	25
3.3.8 Checkout and Orders.....	25
3.3.9 Reviews and Ratings	26
3.3.10 Partner Application	26
3.3.11 Agency Partner Dashboard.....	26
3.3.12 Collaborator Dashboard.....	26
3.3.13 Admin Partner Approval	26
3.3.14 Admin Marketplace Management.....	26
3.3.15 Admin Analytics and Commission Management.....	27
3.4 Non-Functional Requirements.....	27
3.4.1 Performance Requirements	27
3.4.2 Scalability Requirements.....	27
3.4.3 Security Requirements	27
3.4.4 Usability Requirements	27
3.4.5 Reliability Requirements.....	28
3.4.6 Maintainability Requirements	28

3.4.7 Portability Requirements	28
3.4.8 Data Integrity Requirements	28
3.4.9 Availability Requirements.....	28
3.4.10 Extensibility Requirements	29
3.4.11 Backup and Recovery Requirements	29
3.4.12 Load Handling Requirements.....	29
Chapter 4	30
DESIGN AND ARCHITECTURE	30
4 Design and Architecture.....	30
4.1 System Architecture.....	30
4.1.1 Relationships Between Modules	32
4.2. ER Diagram	33
4.3 Process Flow/Representation	35
4.3.1 Activity Diagrams	37
4.3.1.1 User: AI Trip Generation and PDF Export Activity Diagram.....	37
4.3.1.2 User: Marketplace Purchase Activity Diagram	39
4.3.1.3 Admin: Partner Approval and Marketplace Monitoring Activity Diagram.....	40
4.4 Design Models.....	41
4.4.1 Sequence Diagram.....	41
4.4.1.1 Sequence Diagram for User Core Process	41
4.4.1.2 Sequence Diagram for Partner Core Process.....	43
4.4.1.3 Sequence Diagram for Admin Core Process	45
4.4.2.1 Class Diagram	48
Chapter 5	51
IMPLEMENTATION	51

5 Implementation	52
5.1 Algorithm	52
5.1.1 User Authentication/Login	52
5.1.2 AI Trip Generation	53
5.1.3 AI Trip Re-Optimization	53
5.1.4 Weather and Travel Support	54
5.1.5 PDF Travel Report Export.....	55
5.1.6 Publish Marketplace Listing	56
5.1.7 Dummy Checkout and Order Creation	56
5.1.8 Reviews and Ratings	57
5.1.9 Partner Application Approval	58
5.1.10 Commission and Earnings Tracking.....	58
5.2 External APIs	59
5.3 User Interface	60
5.3.1 Sign-in / Sign-up Screen	61
5.3.2 AI Trip Generator Screen	62
5.3.3 Generated Trip Details Screen.....	64
5.3.4 Marketplace Screen	65
5.3.5 Admin Dashboard Screen	67
Chapter 6	69
TESTING AND EVALUATION	69
6 Testing and Evaluation	70
6.1 Manual Testing.....	70
6.1.1 System Testing.....	70
6.1.2 Unit Testing	71

6.1.3 Functional Testing.....	72
6.1.4 Integration Testing.....	73
6.2 Tools and Technologies.....	74
Chapter 7	77
CONCLUSION AND FUTURE WORK	77
Conclusion.....	78
7.1 Future Work.....	78
Chapter 8	80
REFERENCE	80
References	81

List of Figures

Figure 3.1: Use Case.....	12
Figure 4.1: Itinera System Architecture	32
Figure 4.2: Itinera ER Diagram.....	33
Figure 4.3: User AI Trip Generation and PDF Export Activity Diagram.....	35
Figure 4.4: User Marketplace Purchase Activity Diagram.....	36
Figure 4.5: Admin Partner Approval and Marketplace Monitoring Activity Diagram	37
Figure 4.6: Sequence Diagram for User Core Process	38
Figure 4.7: Sequence Diagram for Partner Core Process	39
Figure 4.8: Sequence Diagram for Admin Core Process.....	40
Figure 4.9: Class Diagram of Itinera System	41
Figure 5.1: Sign-in / Sign-up Screen.....	52
Figure 5.2: AI Trip Generator Screen	52
Figure 5.3: Generated Trip Details Screen	52
Figure 5.4: Marketplace Screen.....	53
Figure 5.5: Admin / Partner Dashboard Screen	53

List of Tables

Table 1.1 Relevance to Course Module.....	2
Table 3.1: User Registration and Login	13
Table 3.2: Generate AI Trip.....	14
Table 3.3: Manage Trips	14
Table 3.4: Re-Optimize Trip	15
Table 3.5: Weather and Travel Support.....	16
Table 3.6: Export Trip PDF	16
Table 3.7: Browse Marketplace	17
Table 3.8: Purchase Package / Plan.....	18
Table 3.9: Reviews and Ratings	18
Table 3.10: Apply as Partner	19
Table 3.11: Publish Travel Package	20
Table 3.12: Publish Paid Travel Plan.....	20
Table 3.13: Manage Orders and Earnings.....	21
Table 3.14: Approve Partners.....	22
Table 3.15: Manage Users	22
Table 3.16: Manage Marketplace	23
Table 3.17: View Analytics and Commission	24
Table 5.1: External APIs	50
Table 6.1: System Testing Table.....	54
Table 6.2: Unit Function Table.....	55
Table 6.3: Functional Testing Table.....	56
Table 6.4: Integration Testing Table.....	58
Table 6.5: Tools and Technologies Table.....	58

Chapter 1

INTRODUCTION

1 Introduction

Itinera is an AI-powered travel planning and marketplace platform developed as a Final Year Project at COMSATS University Islamabad, Sahiwal Campus. The main purpose of the system is to make travel planning easier, faster, and more organized by using artificial intelligence to generate personalized trip plans. Instead of manually searching for hotels, destinations, activities, and schedules from different websites, users can enter basic travel details such as origin, destination, budget, duration, and group size, and the system generates a complete travel itinerary.

The AI-generated itinerary includes hotel suggestions, day-wise activities, travel recommendations, locations, coordinates, and other important trip details. To make the travel experience more useful, Itinera also provides live destination weather forecasts, smart packing suggestions, travel document checklists, and weather-based travel alerts. Users can also export their complete travel plan as a printable PDF report, which includes trip summaries, hotels, activities, essentials, and checklists.

Another important feature of Itinera is the AI trip re-optimizer. This allows users to improve or modify an existing trip plan by giving simple instructions such as making the trip cheaper, reducing walking, adding rainy-day alternatives, making the trip family friendly, or optimizing the plan for food experiences. This makes the platform more flexible and user-centered.

In addition to AI-based trip planning, Itinera includes a travel marketplace where agency partners and collaborators can publish travel packages and paid travel plans. Users can browse these marketplace listings, purchase them through a dummy checkout flow, manage their orders, and submit reviews and ratings. The platform also includes role-based dashboards for Normal Users, Admins, Agency Partners, and Collaborators/Travel Bloggers.

Itinera is developed using modern web technologies including Next.js [1], React, TypeScript, Tailwind CSS, ShadCN UI, Convex, Clerk Authentication, OpenAI SDK, Open-Meteo API, Pexels API, Mapbox, Arcjet, and Resend. These technologies help make the system scalable, responsive, secure, and suitable for real-world travel planning and marketplace operations.

1.1 Relevance to Course Module

The development of Itinera applies concepts learned throughout the BSCS degree program, including software engineering, database systems, web development, artificial intelligence, human-computer interaction, and information security.

Table 1.1 Relevance to Course Module

Course Module	Application in Itinera
Software Engineering	Prepared SRS, SDD, UML diagrams, and followed a structured development approach.
Database Systems	Designed Convex database tables for users, trips, marketplace packages, orders, reviews, and commissions.
Web Technologies	Developed a responsive web application using Next.js, React, TypeScript, and Tailwind CSS.
Artificial Intelligence	Integrated AI for trip generation and trip re-optimization using GPT-4o-mini.
Human-Computer Interaction	Designed user-friendly dashboards and interfaces for users, admins, agencies, and collaborators.
Information Security	Implemented Clerk authentication, role-based access control, Arcjet protection, and secure user sessions.

The table 1.1 outlines how different BSCS course modules are practically applied in the development of Itinera. It shows the connection between academic subjects such as Software Engineering, Database Systems, Web Technologies, Artificial Intelligence, HCI, and Information Security with the actual features and implementation of the project.

1.2 Project Background

Travel planning has become increasingly complex due to the growing number of destinations, hotels, activities, weather conditions, budgets, and user preferences that must be considered before finalizing a trip. Many travelers spend significant time searching across multiple

platforms to compare hotels, attractions, routes, and travel plans. This process is often time-consuming, unorganized, and difficult for users who want quick and personalized recommendations.

Existing travel platforms usually provide separate services such as hotel booking, destination browsing, package listings, or reviews. However, most of them do not offer a complete solution where users can generate personalized itineraries, view weather-based travel support, export trip reports, and access marketplace packages in one system. This creates a need for an intelligent and integrated travel planning platform.

Itinera was developed to address these challenges by providing an AI-powered travel planning and marketplace system. The platform allows users to generate customized trips, receive travel recommendations, access weather-based suggestions, and purchase travel plans or packages from agencies and collaborators. By combining AI trip generation, marketplace features, and role-based dashboards, Itinera offers a modern and user-friendly travel management experience.

1.3 Literature Review

The use of artificial intelligence in travel and tourism platforms has increased significantly in recent years. Many modern travel systems provide hotel suggestions, destination recommendations, package browsing, and user reviews. Platforms such as TripAdvisor, Booking.com, and Google Travel support travelers in decision-making, but they mostly focus on search, comparison, and booking rather than fully personalized AI-generated itineraries.

Research and current trends show that AI can improve travel planning by analyzing user preferences such as budget, destination, duration, group size, and travel interests. AI-based systems can generate customized itineraries, recommend suitable activities, and help users make faster travel decisions. Weather forecasting and location-based services also play an important role in improving travel safety and convenience.

Despite these advancements, many travel platforms still lack a unified system that combines AI itinerary generation, weather support, smart packing suggestions, PDF reports, marketplace purchasing, reviews, and partner earnings management. This gap highlights the need for a complete travel planning system that supports both travelers and travel service providers.

1.4 Analysis from Literature Review

The literature review shows that artificial intelligence can make travel planning more personalized, efficient, and user-friendly. Existing platforms provide useful services, but most of them are limited to hotel searching, destination reviews, or package booking. They do not fully support AI-based trip creation, re-optimization, travel checklists, weather alerts, and marketplace management within a single platform.

This analysis highlights the need for an integrated travel planning system that combines AI assistance, real-time travel support, marketplace features, and role-based management. Such a system can reduce manual planning effort for users while also providing agencies and collaborators with a platform to publish and sell travel services.

1.4.1 Key Findings

The review identifies a gap between traditional travel platforms and modern AI-assisted travel planning systems. Users expect faster, personalized, and organized trip planning experiences. Features such as AI-generated itineraries, weather-based suggestions, packing checklists, PDF reports, and trip re-optimization can improve the overall travel experience.

It was also found that marketplace-based travel systems can benefit both users and service providers. Agencies and collaborators can publish travel packages or paid plans, while users can browse, purchase, and review them. Role-based dashboards, secure authentication [2], and commission tracking further improve system management and transparency.

1.4.2 Conceptualization of Itinera

Based on these findings, Itinera was conceptualized as a complete AI-powered travel planning and marketplace platform. Instead of working only as a basic trip planner, Itinera combines AI itinerary generation, trip re-optimization, weather support, smart packing suggestions, travel document checklists, PDF report export, marketplace browsing, dummy checkout, reviews, and commission-based earnings.

The platform provides separate panels for Normal Users, Admins, Agency Partners, and Collaborators/Travel Bloggers. Each role is designed to handle specific operations such as trip generation, package publishing, order management, partner approval, analytics, and earnings

tracking. This approach makes Itinera a scalable, practical, and user-centered solution for modern travel planning.

1.5 Methodology and Software Lifecycle

The Agile Software Development Lifecycle (SDLC) was adopted for the Itinera project to support flexible, iterative, and feature-based development. Agile was selected because the system includes multiple interconnected modules such as AI trip generation, marketplace management, authentication, weather support, PDF report export, partner dashboards, and order handling. The iterative approach allowed each module to be developed, tested, and improved step by step.

1.5.1 Key Phases

- Requirement Gathering:

Functional and non-functional requirements were identified, including user registration, AI trip generation, trip management, marketplace browsing, partner package publishing, dummy checkout, reviews, and admin management.

- System Design:

The system architecture, database structure, UML diagrams, and user interface layouts were designed. Separate role-based panels were planned for Normal Users, Admins, Agency Partners, and Collaborators.

- Implementation:

The frontend was developed using Next.js 15, React, TypeScript, Tailwind CSS, and ShadCN UI. Backend functions and database operations were implemented using Convex and Next.js API routes.

- Integration of Services:

External services were integrated, including Clerk Authentication, OpenAI [3] SDK for AI trip generation and re-optimization, Open-Meteo API for weather, Pexels API for images, Mapbox for maps, Arcjet for protection, and resend for email.

- Testing and Validation:

Unit testing, functional testing, integration testing, and system testing were performed to verify system performance, security, and correct working of all major modules

- **Deployment:**

The web application was prepared for deployment on a cloud-based platform to ensure accessibility, scalability, and smooth user experience.

- **Maintenance:**

Bug fixing, performance improvement, feature updates, and security monitoring were considered for long-term reliability and future enhancement.

1.5.2 Rationale behind Selected Methodology

Agile methodology was selected because Itinera required continuous development, testing, and improvement of multiple modules. Features such as AI trip generation, marketplace orders, partner dashboards, weather support, and trip re-optimization could be developed independently and improved through feedback. Compared to the traditional waterfall model, Agile provided more flexibility in handling changes and integrating new services. This approach helped ensure that the final system remained scalable, user-friendly, and aligned with the goals of an AI-based travel planning and marketplace platform.

Chapter 2

PROBLEM DEFINITION

2 Problem Definition

The travel and tourism industry is becoming increasingly digital, with users expecting fast, personalized, and convenient travel planning solutions. However, many existing travel platforms still provide limited and scattered services such as hotel searching, package browsing, or destination reviews. Users often have to visit multiple websites to plan a complete trip, compare options, check weather conditions, prepare travel essentials, and manage bookings.

The Itinera project was developed to address these challenges by introducing an AI-powered travel planning and marketplace platform. The system helps users generate personalized travel itineraries, receive weather-based travel support, export printable travel reports, and browse marketplace packages from agencies and collaborators. This chapter defines the main problem, outlines the project deliverables, and discusses the limitations of existing travel planning systems that Itinera aims to improve.

2.1 Problem Statement

Traditional travel planning platforms are often limited to basic hotel booking, destination searching, or package listing. These systems usually do not provide a complete personalized itinerary based on user preferences such as budget, duration, group size, origin, and destination. As a result, users spend extra time manually searching, comparing, and organizing their travel plans.

A deeper analysis shows several major issues in current systems. Many platforms lack AI-based trip generation, smart trip re-optimization, weather-based alerts, packing suggestions, and travel document checklists. Users also face difficulty in keeping all travel information organized in one place. In addition, most systems do not provide printable PDF reports containing complete trip details.

Existing travel marketplaces also have limitations in managing agency packages, collaborator travel plans, reviews, orders, earnings, and commissions in a unified system. Admins may face challenges in approving partners, monitoring marketplace activity, and tracking revenue

analytics. These limitations create a need for a centralized travel platform that combines AI planning, marketplace management, and role-based dashboards.

Itinera addresses these challenges by providing a secure, intelligent, and user-friendly travel planning system. The platform enables AI itinerary generation, trip re-optimization, live weather support, PDF travel reports, marketplace browsing, dummy checkout, reviews, commission tracking, and dedicated dashboards for users, admins, agencies, and collaborators.

2.2 Deliverables and Development Requirements

The Itinera project aims to deliver a complete AI-powered travel planning and marketplace web application with role-based access, trip generation, travel support, and marketplace management features. The expected deliverables and development requirements are listed below:

2.2.1 Major Deliverables:

- **AI Trip Planning Web Application**
Enabling users to generate personalized trips based on origin, destination, budget, duration, and group size.
- **User Panel**
Allowing users to generate trips, view itineraries, manage bookings, export PDF reports, and review marketplace plans.
- **Admin Dashboard**
Enabling admins to manage users, approve partner applications, monitor orders, reviews, revenue, and platform activity.
- **Agency Partner Dashboard**
Allowing agencies to publish travel packages, manage orders, and track earnings.
- **Collaborator Dashboard**
Enabling travel bloggers or collaborators to publish paid travel plans and monitor sales.
- **AI Trip Re-Optimize**
Allowing users to improve existing trips using instructions such as reducing cost, adding rainy-day alternatives, or making the trip family friendly.
- **Weather and Travel Support**

Providing live weather forecasts, travel alerts, packing suggestions, and document checklists.

- Marketplace and Checkout System
Enabling users to browse packages, purchase plans through dummy checkout, submit reviews, and manage orders.
- Technical Documentation and Source Code Repository
Ensuring maintainability, clarity, and future development support.

2.2.2 Development Requirements

- Frontend: Next.js 15, React, TypeScript, Tailwind CSS, ShadCN [9] UI, Lucide React Icons
- Backend: Convex, Next.js API Routes
- Database: Convex Database
- Authentication: Clerk Authentication with role-based access
- AI Integration: OpenAI SDK, GitHub Models / Azure AI Inference, GPT-4o-mini
- External APIs: Open-Meteo API, Pexels [4] API, Mapbox
- Security and Email: Arcjet, Resend
- Deployment: Vercel
- Tools: Git, Visual Studio Code

2.3 Current System

In the current scenario, many travel planning platforms provide limited and separate services such as hotel booking, destination searching, travel package listing, or user reviews. Most of these systems do not provide a complete AI-based solution for generating personalized itineraries according to user preferences such as budget, duration, group size, origin, and destination.

Users often have to visit multiple websites to search for hotels, activities, weather conditions, locations, and travel packages. This makes the planning process time-consuming and

unorganized. Many existing platforms also lack smart travel support features such as packing suggestions, travel document checklists, weather alerts, and printable trip reports.

Another limitation of current systems is the lack of AI-based trip re-optimization. Once a trip plan is created, users usually cannot easily modify it according to preferences such as reducing cost, adding rainy-day alternatives, minimizing walking, or making the trip family friendly. Similarly, many travel marketplaces do not provide proper role-based dashboards for agencies, collaborators, and admins to manage packages, orders, earnings, reviews, and commissions efficiently.

Itinera overcomes these limitations by providing an integrated AI-powered travel planning and marketplace platform. The system offers AI trip generation, trip re-optimization, live weather support, PDF travel reports, marketplace browsing, dummy checkout, reviews, commission tracking, and role-based dashboards for users, admins, agencies, and collaborators. This makes travel planning more personalized, organized, and efficient.

Chapter 3

REQUIREMENTS ANALYSIS

3 Requirement Analysis

Requirement Analysis, also known as Requirement Engineering, is the process of identifying, analyzing, documenting, and validating the needs of a software system. It helps define what the system should do, how it should behave, and what conditions must be fulfilled to meet user expectations and project objectives. This phase is important because it provides a clear foundation for system design, development, testing, and final evaluation.

For Itinera, requirement analysis focuses on understanding the needs of different users, including Normal Users, Admins, Agency Partners, and Collaborators/Travel Bloggers. Each user role has different responsibilities and interactions within the platform. Normal users require features for generating AI-based trips, managing itineraries, viewing weather support, exporting PDF reports, browsing marketplace listings, purchasing plans, and submitting reviews. Agency partners and collaborators require dashboards to publish travel packages or paid travel plans, manage orders, and track earnings. Admins require control over users, partner approvals, marketplace monitoring, analytics, and commission management.

The main requirements of Itinera include AI trip generation, trip re-optimization, trip management, weather-based travel support, PDF travel report export, marketplace browsing, dummy checkout, order handling, reviews and ratings, partner package publishing, collaborator plan publishing, and admin monitoring. By clearly defining these requirements, the development process becomes more organized and reduces the chances of missing important functionality. This phase ensures that the system features are properly planned before moving toward design and implementation.

3.1 Use Case Diagram

This Figure 3.1 shows the main interactions between users and the Itinera system. It includes core functions such as AI trip generation, marketplace browsing, package management, partner approval, and admin monitoring.

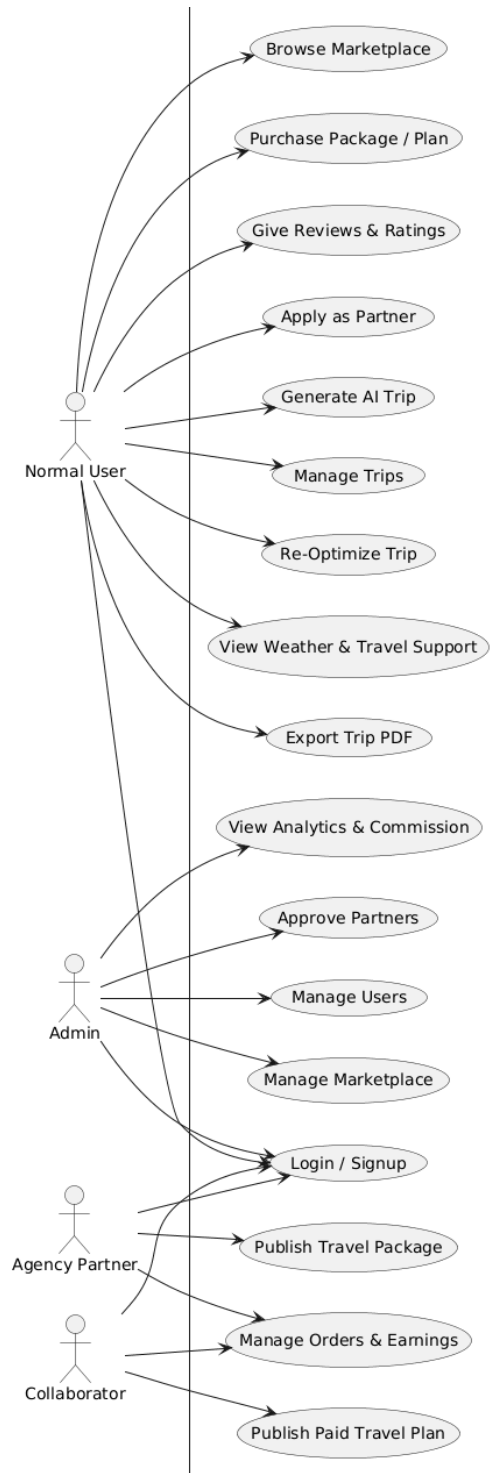


Figure 3.1: Use Case

3.2 Use Case Description

The following use case descriptions are prepared according to the Itinera use case diagram and follow the same report format as your provided sample.

Table 3.1: User Registration and Login

Field	Description
Use Case ID	UC#1
Use Case Name	Login / Signup
Description	Allows users, admins, agency partners, and collaborators to register or log in to the Itinera system.
Primary Actor	Normal User / Admin / Agency Partner / Collaborator
Secondary Actor	Clerk Authentication
Pre-Condition	User must have internet access and a valid email address.
Post-Condition	User is authenticated and redirected to the relevant dashboard.
Basic Flow	1. User opens login/signup page. 2. User enters credentials. 3. System verifies credentials through Clerk. 4. User is redirected to the dashboard.
Alternate Flow	Invalid credentials → System displays an error message.

Table 3.2: Generate AI Trip

Field	Description
Use Case ID	UC#2
Use Case Name	Generate AI Trip
Description	Allows users to generate a personalized travel itinerary using AI.
Primary Actor	Normal User
Secondary Actor	AI Model
Pre-Condition	User must be logged in.
Post-Condition	AI-generated trip is created and saved in the system.
Basic Flow	1. User enters origin, destination, budget, duration, and group size. 2. System sends data to AI model. 3. AI generates hotels, itinerary, activities, and recommendations. 4. Trip is displayed to the user.
Alternate Flow	Invalid or missing input → System asks user to correct the details.

Table 3.3: Manage Trips

Field	Description
Use Case ID	UC#3
Use Case Name	Manage Trips
Description	Allows users to view, save, and manage their generated trips.
Primary Actor	Normal User

Secondary Actor	None
Pre-Condition	User must be logged in and have generated trips.
Post-Condition	User can view or manage saved trip records.
Basic Flow	1. User opens trip dashboard. 2. System displays saved trips. 3. User selects a trip to view details.
Alternate Flow	No trips found → System displays an empty trip message.

Table 3.4: Re-Optimize Trip

Field	Description
Use Case ID	UC#4
Use Case Name	Re-Optimize Trip
Description	Allows users to improve an existing trip using AI instructions.
Primary Actor	Normal User
Secondary Actor	AI Model
Pre-Condition	User must have an existing trip.
Post-Condition	Updated itinerary is generated according to user instructions.
Basic Flow	1. User selects a saved trip. 2. User enters instruction such as make cheaper or reduce walking. 3. System sends request to AI model. 4. Updated trip plan is displayed.
Alternate Flow	AI response fails → System shows an error and allows retry.

Table 3.5: Weather and Travel Support

Field	Description
Use Case ID	UC#5
Use Case Name	View Weather and Travel Support
Description	Provides live weather forecast, travel alerts, packing suggestions, and document checklist.
Primary Actor	Normal User
Secondary Actor	Open-Meteo API
Pre-Condition	User must select or generate a destination-based trip.
Post-Condition	Weather and travel support information is displayed.
Basic Flow	1. User opens trip details. 2. System fetches destination weather. 3. System displays forecast, alerts, packing list, and documents checklist.
Alternate Flow	Weather data unavailable → System displays default travel suggestions.

Table 3.6: Export Trip PDF

Field	Description
Use Case ID	UC#6
Use Case Name	Export Trip PDF
Description	Allows users to download a printable travel report.
Primary Actor	Normal User

Secondary Actor	None
Pre-Condition	Trip must be generated or saved.
Post-Condition	PDF report is created and downloaded.
Basic Flow	1. User opens trip details. 2. User clicks export PDF. 3. System generates report with itinerary, hotels, activities, and checklists. 4. PDF is downloaded.
Alternate Flow	PDF generation fails → System displays error message.

Table 3.7: Browse Marketplace

Field	Description
Use Case ID	UC#7
Use Case Name	Browse Marketplace
Description	Allows users to browse travel packages and paid travel plans.
Primary Actor	Normal User
Secondary Actor	Agency Partner / Collaborator
Pre-Condition	Marketplace listings must be available.
Post-Condition	User views available packages and plans.
Basic Flow	1. User opens marketplace. 2. System displays travel packages and plans. 3. User selects a listing to view details.
Alternate Flow	No listings available → System displays no package message.

Table 3.8: Purchase Package / Plan

Field	Description
Use Case ID	UC#8
Use Case Name	Purchase Package / Plan
Description	Allows users to purchase marketplace packages or paid travel plans through dummy checkout.
Primary Actor	Normal User
Secondary Actor	Agency Partner / Collaborator
Pre-Condition	User must be logged in and package or plan must be available.
Post-Condition	Order is created and commission is recorded.
Basic Flow	1. User selects package or plan. 2. User proceeds to dummy checkout. 3. System confirms order. 4. Order and commission records are updated.
Alternate Flow	Checkout cancelled → Order is not created.

Table 3.9: Reviews and Ratings

Field	Description
Use Case ID	UC#9
Use Case Name	Give Reviews and Ratings
Description	Allows users to review purchased travel packages or plans.
Primary Actor	Normal User
Secondary Actor	None

Pre-Condition	User must have purchased a package or plan.
Post-Condition	Review and rating are saved in the system.
Basic Flow	1. User opens completed order. 2. User enters rating and review. 3. System saves review. 4. Review appears on marketplace listing.
Alternate Flow	Empty review or invalid rating → System prompts correction.

Table 3.10: Apply as Partner

Field	Description
Use Case ID	UC#10
Use Case Name	Apply as Partner
Description	Allows users to apply as an agency partner or collaborator.
Primary Actor	Normal User
Secondary Actor	Admin
Pre-Condition	User must be logged in.
Post-Condition	Partner application is submitted for admin approval.
Basic Flow	1. User opens partner application page. 2. User selects partner type. 3. User submits required details. 4. Application is sent to admin.
Alternate Flow	Incomplete details → System asks user to complete the form.

Table 3.11: Publish Travel Package

Field	Description
Use Case ID	UC#11
Use Case Name	Publish Travel Package
Description	Allows agency partners to create and publish travel packages.
Primary Actor	Agency Partner
Secondary Actor	Admin
Pre-Condition	Agency partner must be approved and logged in.
Post-Condition	Travel package is published in the marketplace.
Basic Flow	1. Agency opens dashboard. 2. Agency enters package details, price, destination, and images. 3. System saves package. 4. Package appears in marketplace.
Alternate Flow	Invalid details → System displays validation error.

Table 3.12: Publish Paid Travel Plan

Field	Description
Use Case ID	UC#12
Use Case Name	Publish Paid Travel Plan
Description	Allows collaborators or travel bloggers to publish paid travel plans.
Primary Actor	Collaborator
Secondary Actor	Admin
Pre-Condition	Collaborator must be approved and logged in.

Post-Condition	Paid travel plan is published in the marketplace.
Basic Flow	1. Collaborator opens dashboard. 2. Collaborator enters travel plan details and price. 3. System saves the plan. 4. Plan becomes available for users.
Alternate Flow	Missing information → System prompts collaborator to complete details.

Table 3.13: Manage Orders and Earnings

Field	Description
Use Case ID	UC#13
Use Case Name	Manage Orders and Earnings
Description	Allows partners and collaborators to view orders and track earnings.
Primary Actor	Agency Partner / Collaborator
Secondary Actor	None
Pre-Condition	Partner or collaborator must be logged in.
Post-Condition	Orders and earnings are displayed.
Basic Flow	1. Partner opens dashboard. 2. System retrieves orders and earnings. 3. Partner views sales, commission, and revenue details.
Alternate Flow	No orders found → System displays empty order history.

Table 3.14: Approve Partners

Field	Description
Use Case ID	UC#14

Use Case Name	Approve Partners
Description	Allows admin to approve or reject partner applications.
Primary Actor	Admin
Secondary Actor	Agency Partner / Collaborator
Pre-Condition	Partner application must be submitted.
Post-Condition	Application status is updated.
Basic Flow	1. Admin opens partner applications. 2. Admin reviews submitted details. 3. Admin approves or rejects application. 4. System updates user role/status.
Alternate Flow	Invalid application → Admin rejects the request.

Table 3.15: Manage Users

Field	Description
Use Case ID	UC#15
Use Case Name	Manage Users
Description	Allows admin to view and manage platform users.
Primary Actor	Admin
Secondary Actor	None
Pre-Condition	Admin must be logged in.
Post-Condition	User records are updated or managed.
Basic Flow	1. Admin opens user management page. 2. System displays all users. 3. Admin views or manages user details.

Alternate Flow	User not found → System displays no record message.
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Table 3.16: Manage Marketplace

Field	Description
Use Case ID	UC#16
Use Case Name	Manage Marketplace
Description	Allows admin to monitor and manage packages, plans, reviews, and orders.
Primary Actor	Admin
Secondary Actor	Agency Partner / Collaborator
Pre-Condition	Admin must be logged in.
Post-Condition	Marketplace records are managed successfully.
Basic Flow	1. Admin opens marketplace management. 2. System displays listings, orders, and reviews. 3. Admin reviews or manages marketplace data.
Alternate Flow	Invalid action → System displays error message.

Table 3.17: View Analytics and Commission

Field	Description
Use Case ID	UC#17
Use Case Name	View Analytics and Commission
Description	Allows admin to view revenue analytics and commission records.
Primary Actor	Admin

Secondary Actor	None
Pre-Condition	Orders must exist in the system.
Post-Condition	Revenue, commission, and marketplace statistics are displayed.
Basic Flow	1. Admin opens analytics dashboard. 2. System calculates orders, earnings, and commission. 3. Admin views analytics reports.
Alternate Flow	No revenue data found → System displays empty analytics view.

3.3 Functional Requirements

Functional requirements define the specific operations and services that the Itinera platform must perform to achieve its intended objectives. These requirements describe how users, admins, agency partners, and collaborators interact with the system to perform travel planning, marketplace, booking, and management tasks. The following subsections outline the major functional requirements of the Itinera system, adapted according to the provided report format.

3.3.1 User Login and Signup

The system shall allow users to create an account and log in to access Itinera. Users can register using valid credentials through Clerk Authentication. After successful login, the system redirects users to their respective dashboard based on their role, such as Normal User, Admin, Agency Partner, or Collaborator. If invalid credentials are entered, the system displays an appropriate error message.

3.3.2 AI Trip Generation

The system shall allow users to generate personalized trips by entering origin, destination, budget, duration, and group size. The AI model processes the provided information and generates hotels, day-wise itinerary, activities, travel recommendations, locations, and coordinates. The generated trip is displayed to the user and saved in the system.

3.3.3 Trip Management

The system shall allow users to view and manage their generated trips. Users can open saved trips, review itinerary details, view hotel suggestions, check activities, and manage their travel plans from the user dashboard. If no trip is available, the system displays an empty trip message.

3.3.4 Weather and Travel Support

The system shall provide live destination weather forecasts using the Open-Meteo API. Based on weather conditions, the system also provides travel alerts, smart packing suggestions, and travel document checklists. This helps users prepare for their trips more efficiently.

3.3.5 PDF Travel Report

The system shall allow users to export their complete travel plan as a printable PDF report. The report includes trip summary, hotels, itinerary, activities, travel essentials, packing suggestions, and document checklists. If PDF generation fails, the system displays an error message.

3.3.6 AI Trip Re-Optimization

The system shall allow users to improve existing trips using AI-based instructions. Users can request changes such as making the trip cheaper, reducing walking, adding rainy-day alternatives, making the trip family friendly, or optimizing it for food experiences. The AI model updates the itinerary according to the user's instruction.

3.3.7 Marketplace Browsing

The system shall allow users to browse marketplace listings published by agency partners and collaborators. Users can view travel packages, paid travel plans, prices, descriptions, destinations, images, and other relevant details. If no listings are available, the system displays an appropriate message.

3.3.8 Checkout and Orders

The system shall allow users to purchase travel packages or paid travel plans through a dummy checkout flow. After checkout completion, the system creates an order record and updates the related marketplace and commission data. Users can also view and manage their order history.

3.3.9 Reviews and Ratings

The system shall allow users to submit reviews and ratings for purchased travel packages or plans. Reviews are saved in the database and displayed on the relevant marketplace listing. The system must validate review content and rating values before submission.

3.3.10 Partner Application

The system shall allow users to apply as an Agency Partner or Collaborator/Travel Blogger. Applicants submit required details, and the application is forwarded to the admin for approval. Once approved, the user gains access to the relevant partner dashboard.

3.3.11 Agency Partner Dashboard

The system shall provide agency partners with a dashboard to publish travel packages, manage published listings, view orders, and track earnings. Agency partners can add package details such as title, destination, price, description, and images.

3.3.12 Collaborator Dashboard

The system shall provide collaborators or travel bloggers with a dashboard to publish paid travel plans, manage listings, view sales, and track earnings. Collaborators can create travel plans and offer them to users through the marketplace.

3.3.13 Admin Partner Approval

The system shall allow admins to review, approve, or reject partner applications. Admins can verify submitted details and update the applicant's role or application status accordingly.

3.3.14 Admin Marketplace Management

The system shall allow admins to monitor and manage marketplace packages, paid travel plans, orders, reviews, and partner activity. This ensures proper control over marketplace content and platform operations.

3.3.15 Admin Analytics and Commission Management

The system shall allow admins to view revenue analytics, order statistics, partner earnings, and commission records. The commission ledger keeps track of platform earnings from marketplace orders and supports transparent financial monitoring.

3.4 Non-Functional Requirements

Non-functional requirements define the quality standards that the Itinera platform must meet to ensure secure, reliable, efficient, and user-friendly performance. These requirements describe how the system should operate rather than what functions it performs. The following requirements are adapted according to the provided format.

3.4.1 Performance Requirements

The Itinera platform must provide fast response times for major operations such as AI trip generation, marketplace browsing, dashboard loading, weather fetching, and PDF report generation. Under normal conditions, user actions should respond within a few seconds. Backend functions and database queries must be optimized to handle multiple users efficiently.

3.4.2 Scalability Requirements

The system must be scalable to support future growth in users, trips, marketplace packages, orders, reviews, and partner accounts. The use of Next.js, Convex [5], and cloud deployment allows the platform to handle increasing data and user traffic without major changes in architecture.

3.4.3 Security Requirements

Itinera must ensure secure authentication and role-based access control using Clerk Authentication. Only authorized users should access their respective dashboards. Admin, agency partner, collaborator, and normal user permissions must be properly separated. Arcjet should be used for protection against abuse, suspicious requests, and basic security threats.

3.4.4 Usability Requirements

The platform interface must be simple, responsive, and easy to use for all roles. Users should be able to generate trips, browse packages, purchase plans, and export reports without

confusion. Dashboards for admins, agencies, and collaborators should provide clear navigation and meaningful feedback for every action.

3.4.5 Reliability Requirements

The system must remain stable during normal usage and handle errors gracefully. If external services such as AI, weather, image, or map APIs fail, the system should display proper error messages or fallback information instead of crashing.

3.4.6 Maintainability Requirements

The system must be modular and easy to maintain. Separate modules such as authentication, AI trip generation, marketplace, orders, reviews, admin dashboard, and partner dashboards should be developed independently. Clean code structure, proper naming, and Git version control should be followed.

3.4.7 Portability Requirements

Itinera must be accessible through modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The application should work properly on desktops, laptops, tablets, and mobile devices using responsive design.

3.4.8 Data Integrity Requirements

The system must maintain accurate and consistent data for users, trips, partner applications, packages, orders, reviews, and commission records. Validation should be applied to prevent incomplete entries, duplicate records, and incorrect marketplace or trip data.

3.4.9 Availability Requirements

The platform should be available online through reliable cloud deployment. Users, admins, agencies, and collaborators should be able to access their dashboards and system features whenever required, with minimum downtime.

3.4.10 Extensibility Requirements

The system architecture must support future improvements such as real payment gateway integration, advanced AI recommendations, hotel booking integration, flight booking, real-time chat, and enhanced analytics without major restructuring.

3.4.11 Backup and Recovery Requirements

Important system data such as user records, trips, marketplace listings, orders, reviews, and commission ledger should be backed up regularly. In case of system failure, data should be recoverable with minimum loss.

3.4.12 Load Handling Requirements

The Itinera platform should be capable of handling multiple users at the same time, including users generating trips, browsing the marketplace, placing orders, and accessing dashboards. Performance testing should be performed to ensure the system remains stable under concurrent usage.

Chapter 4

DESIGN AND ARCHITECTURE

4 Design and Architecture

In the Design and Architecture phase of the Itinera system, the overall structure of the application is defined along with the interaction between its major components. This phase focuses on designing a user-friendly interface for normal users, admins, agency partners, and collaborators, while also creating a scalable backend structure for AI trip generation, marketplace management, orders, reviews, weather support, and role-based dashboards.

The system is designed to ensure smooth communication between frontend, backend, database, authentication services, AI model, and external APIs. Itinera follows a modular architecture, which makes the platform easier to maintain, extend, and test. Each module is responsible for a specific function, such as trip planning, marketplace handling, partner management, or admin monitoring.

4.1 System Architecture

- Modular Program Structure

Itinera is organized into multiple interconnected modules, where each module performs a specific responsibility within the system. This modular structure allows different parts of the platform to be developed and maintained independently. For example, AI trip generation, marketplace operations, partner dashboards, weather support, and admin analytics can be updated without affecting the entire system.

The modular architecture also supports clear data flow between users, partners, collaborators, and admins. When a user generates a trip, the request is processed through the AI trip module, relevant weather and location data are fetched through external APIs, and the final itinerary is saved in the database. Similarly, when a user purchases a marketplace plan, the order module records the purchase, updates commission data, and displays earnings in the partner dashboard.

- User Authentication Module:
Manages login, signup, session handling, and role-based access using Clerk Authentication.
- AI Trip Generation Module:

Generates personalized travel itineraries based on origin, destination, budget, duration, and group size using GPT-4-mini.

- **Trip Management Module:**
Allows users to view, save, manage, and access previously generated trips.
- **AI Trip Re-Optimization Module:**
Enables users to modify existing trips using AI instructions such as reducing cost, adding rainy-day alternatives, or making the trip family friendly.
- **Weather and Travel Support Module:**
Provides live weather forecasts, travel alerts, smart packing suggestions, and travel document checklists using Open-Meteo API.
- **Marketplace Module:**
Allows users to browse agency travel packages and collaborator paid travel plans.
- **Order and Checkout Module:**
Handles dummy checkout, order creation, booking records, and purchase history.
- **Review and Rating Module:**
Allows users to submit reviews and ratings for purchased packages or plans.
- **Partner Management Module:**
Enables agencies and collaborators to publish listings, manage orders, and track earnings.
- **Admin Management Module:**
Allows admins to approve partner applications, manage users, monitor marketplace activity, view analytics, and manage commission records.
- **Database Module:**
Stores users, trips, partner applications, packages, paid plans, orders, reviews, and commission ledger using convex database.
- **External API Integration Module:**
Integrates Open-Meteo for weather, Pexels for travel images, Mapbox for maps and coordinates, Arcjet for protection, and resend for email services.

4.1.1 Relationships Between Modules

- Normal Users interact with the Authentication, AI Trip Generation, Trip Management, Marketplace, Checkout, Review, and PDF Report modules.
- Agency Partners interact with the Authentication, Package Publishing, Order Management, and Earnings modules.
- Collaborators interact with the Authentication, Paid Travel Plan, Order Management, and Earnings modules.
- Admin interacts with the Admin Management, Partner Approval, Marketplace Monitoring, Analytics, and Commission modules.
- The AI Trip Module connects user travel inputs with the AI model to generate personalized itineraries.
- The Marketplace Module connects users with agency packages and collaborator travel plans.
- The Convex Database acts as the shared data layer for users, trips, orders, reviews, packages, and commission records.

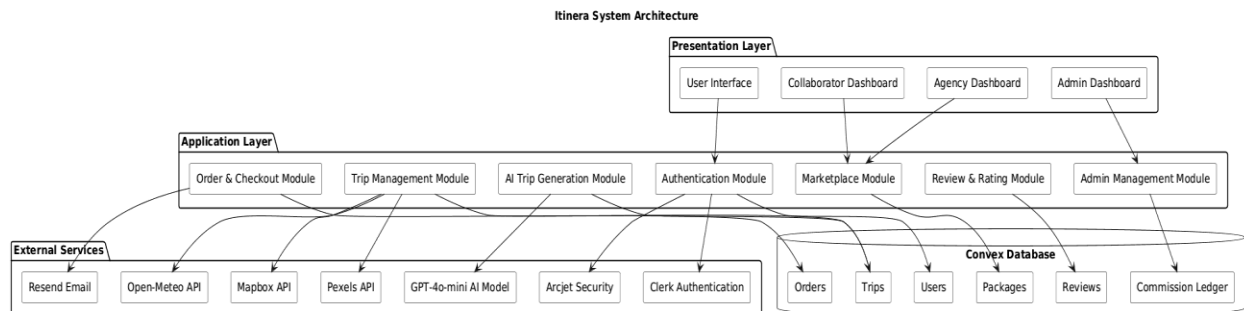


Figure 4.1: Itinera System Architecture

The diagram shows the overall architecture of the Itinera platform using four main parts: Presentation Layer, Application Layer, External Services, and Database Layer. It illustrates how different user panels interact with core modules, while APIs and the Convex database support authentication, AI trip generation, marketplace operations, and data storage.

4.2. ER Diagram

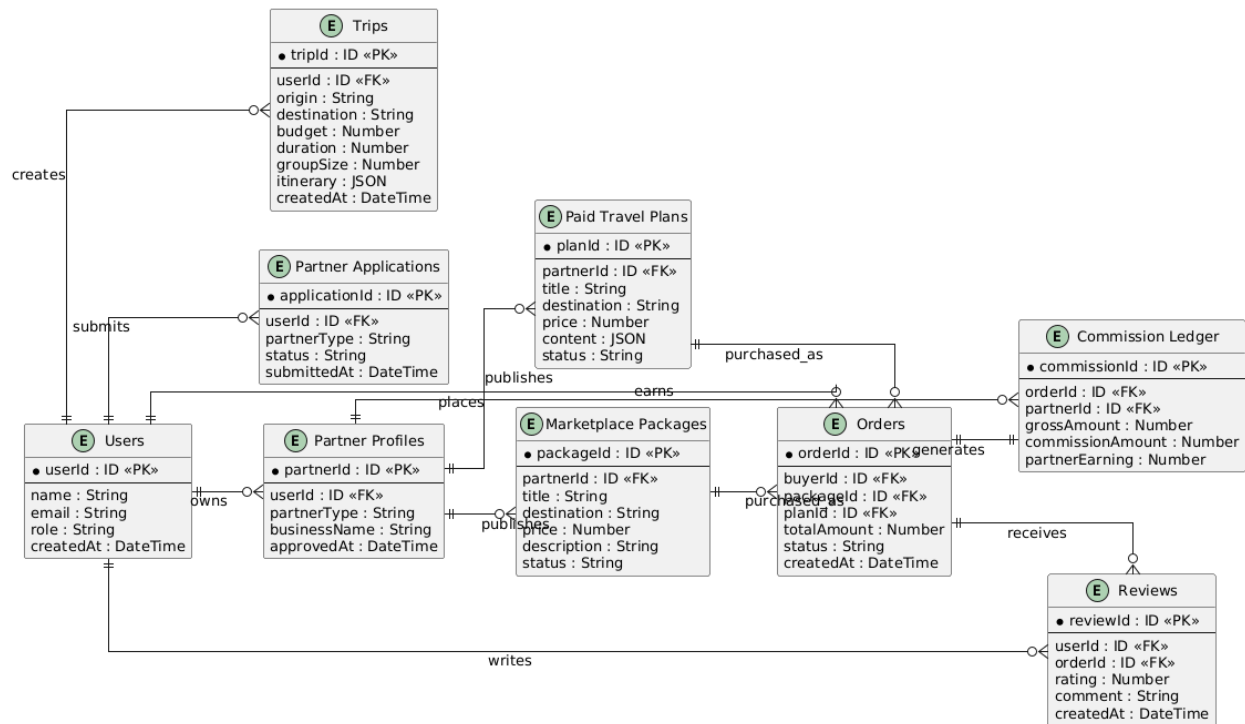


Figure 4.2: Itinera ER Diagram

The figure 4.2 shows the main database entities of the Itinera platform and their relationships. It includes users, trips, partner applications, marketplace packages, paid travel plans, orders, reviews, and commission records.

Key Database Entities and Relational Architecture:

- **User Identity and Partner Profiling:**
 - **Users Entity:** Acts as the central node for identity management, storing fundamental attributes such as name, email, and role. It uses `userId` as the Primary Key (PK).
 - **Partner Applications & Partner Profiles:** A User can *submit* multiple applications (1-to-Many). Upon admin approval, a user *owns* a Partner Profile (1-to-1 relationship), which stores specialized attributes like

businessName and uses userId as a Foreign Key (FK) to link back to the core identity.

- **AI Trip Generation Data:**

- **Trips Entity:** Represents the core AI output. A single user *creates* multiple trips (1-to-Many). This table captures the user's input parameters (origin, destination, budget, duration, groupSize) and stores the complex AI-generated output within a flexible itinerary (JSON) field. It links back to the user via the userId FK.

- **Marketplace Inventory Management:**

- Approved partners (via partnerId FK) can *publish* multiple listings (1-to-Many). The schema separates these offerings into two distinct entities:
 - **Marketplace Packages:** Designed for agency offerings, storing price and description.
 - **Paid Travel Plans:** Designed for collaborators, storing specialized content in a JSON format.

- **Transactional E-Commerce Engine:**

- **Orders Entity:** Serves as the junction for all financial transactions. A user *places* multiple orders (1-to-Many). The order records the buyerId, totalAmount, and links to the specific purchased entity using either packageId or planId as Foreign Keys.

- **Financial Splitting and Feedback Loops:**

- **Commission Ledger:** Every successful order *generates* a corresponding ledger entry (1-to-1 relationship). This table automatically records the grossAmount, the platform's commissionAmount, and the partnerEarning, ensuring transparent financial tracking.
- **Reviews Entity:** To maintain marketplace authenticity, a review is strictly tied to a completed order (an order *receives* a review) and the user who *writes* it. It

records the rating and comment, utilizing orderId and userId as Foreign Keys to prevent unverified feedback.

4.3 Process Flow/Representation

The Itinera platform enables users to plan and manage trips through a highly structured, modular, and user-friendly workflow. The journey begins with a secure, role-based authentication process, ensuring that every individual is seamlessly directed to their appropriate workspace based on their system privileges.

Once authenticated, Normal Users can initiate the core planning process by entering specific travel details, such as origin, destination, budget, duration, and group size. The system feeds this data into the AI model to generate comprehensive, personalized travel itineraries. To enrich the travel experience, users can interact with these itineraries to view live, destination-specific weather support, dynamically re-optimize their trips using natural language instructions, and export their finalized travel reports as printable PDFs for offline use. Beyond personalized planning, users are seamlessly integrated into the platform's e-commerce ecosystem, where they can browse marketplace packages or expert paid travel plans. From there, they can execute purchases via a dummy checkout flow, manage their order history, and submit verified reviews and ratings to help maintain marketplace quality.

On the supply side, the workflow strictly relies on administrative validation. Agency Partners and Collaborators can only access their specialized management dashboards after their initial applications have been formally approved by the system administrators. Once authorized, these partners are empowered to publish detailed travel packages or paid travel plans directly to the public marketplace, manage their active and inactive listings, view incoming customer orders, and transparently track their financial earnings generated from user purchases.

To maintain system integrity and oversee global operations, Administrators govern the overall platform from a centralized control panel. Their operational workflow includes rigorously reviewing and approving or rejecting partner applications, monitoring live marketplace activity to ensure content quality, and managing registered users across all roles. Furthermore, the administrative process includes reviewing transactional orders and utilizing a built-in commission ledger to track platform revenue, ensuring that every dummy checkout correctly

splits the financial earnings between the Itinera platform and the respective service providers. This interconnected process flow guarantees a cohesive, secure, and scalable experience for all actors within the ecosystem.

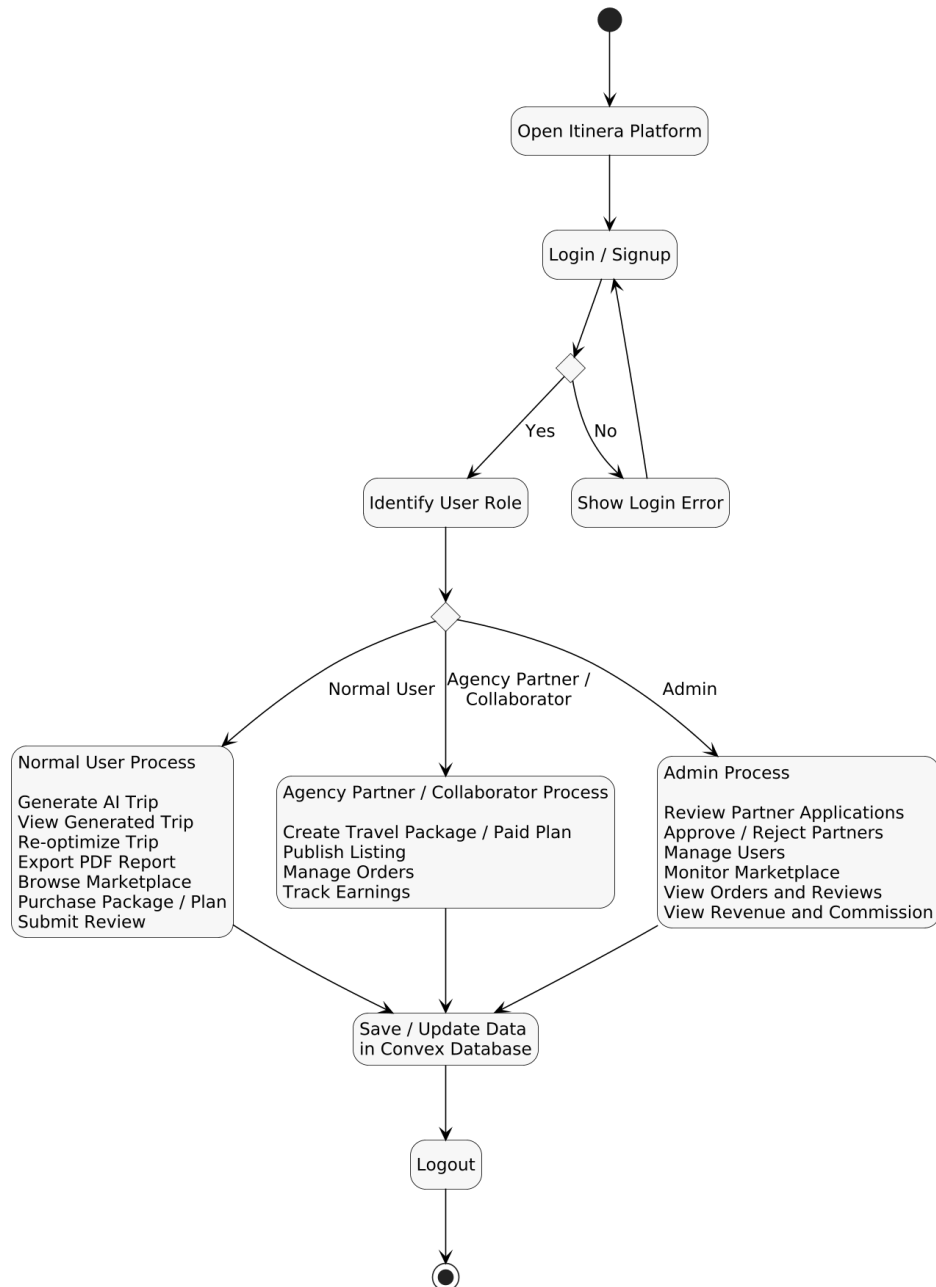


Figure 4.3 Process flow

Figure 4.3 shows the flow of Itinera integrates AI trip planning, travel support, marketplace management, and administrative control into one unified platform, ensuring a smooth and organized travel planning experience.

4.3.1 Activity Diagrams

Activity diagrams show the step-by-step workflow of the major processes performed in the Itinera system. These diagrams are used to represent how different actors move through the system, what actions are performed at each stage, and how the system responds to different conditions. They help in understanding the internal logic of the platform before implementation and make the process flow easier to explain in the report.

In Itinera, activity diagrams describe the interaction of Normal Users, Admins, Agency Partners, and Collaborators with different modules of the system. They show how users log in, generate AI-based travel plans, view trip details, export PDF reports, browse marketplace listings, and complete dummy checkout. They also represent administrative workflows such as partner approval, marketplace monitoring, order tracking, and commission management.

These diagrams also highlight important decision points in the system, such as whether login credentials are valid, whether trip details are complete, whether AI trip generation is successful, whether the user wants to export a PDF, whether checkout is confirmed, and whether a partner application should be approved or rejected. By showing these conditions clearly, activity diagrams reduce ambiguity and help ensure that the system workflow is logical, complete, and easy to understand.

- Major Process Flow:
- User generating AI trip and exporting PDF report.
- User browsing marketplace and purchasing package/plan.
- Admin approving partners and monitoring marketplace activity.

4.3.1.1 User: AI Trip Generation and PDF Export Activity Diagram

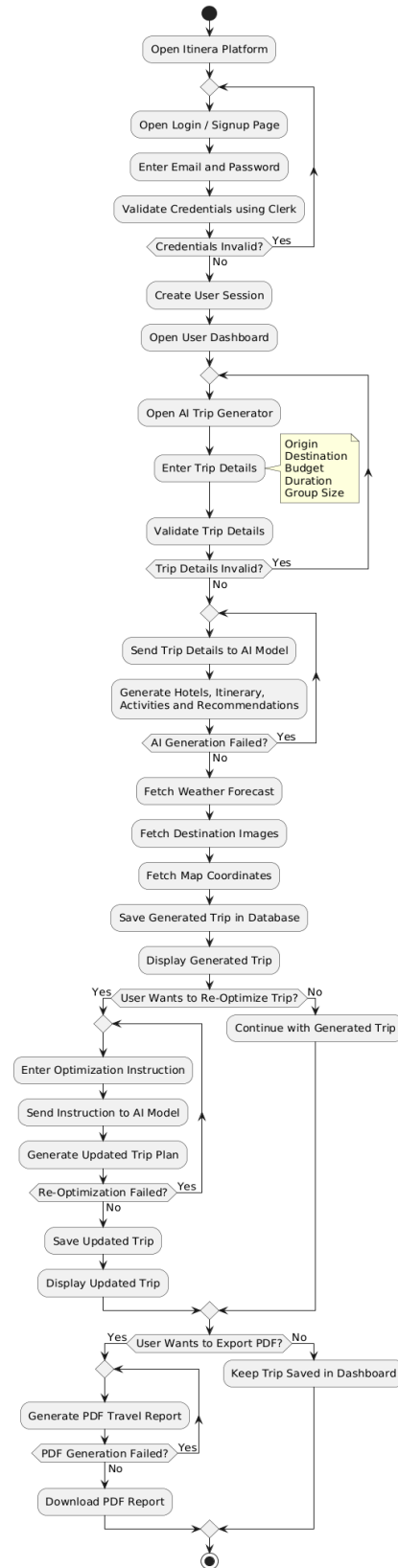


Figure 4.3: User AI Trip Generation and PDF Export Activity Diagram

This figure 4.3 shows the correct workflow for generating an AI-based trip in Itinera. The user cannot access the dashboard until credentials are valid, and invalid trip details return to the input form before AI generation begins.

4.3.1.2 User: Marketplace Purchase Activity Diagram

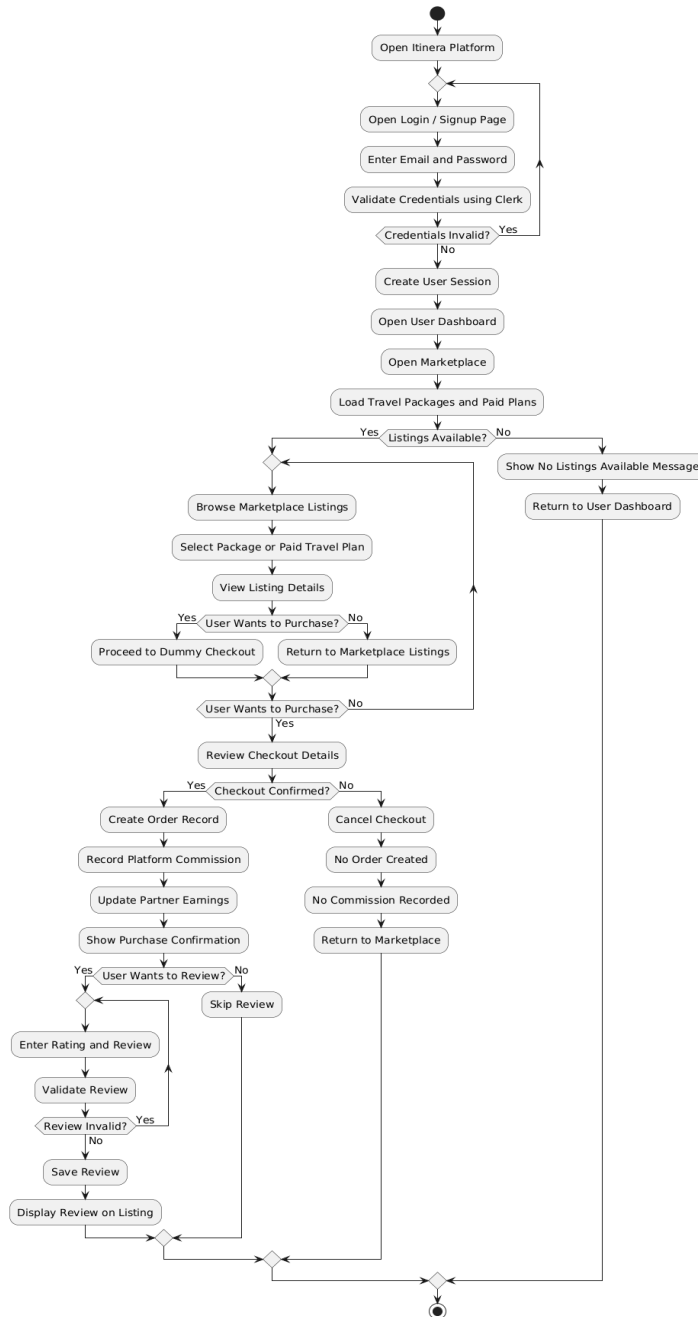


Figure 4.4: User Marketplace Purchase Activity Diagram

This figure 4.4 explains the purchase workflow of Itinera’s marketplace. It shows that only confirmed checkout creates an order, records commission, and updates partner earnings, while cancelled checkout returns the user without creating any transaction.

4.3.1.3 Admin: Partner Approval and Marketplace Monitoring Activity Diagram



Figure 4.5: Admin Partner Approval and Marketplace Monitoring Activity Diagram

4.4 Design Models

Design models represent the internal working, communication flow, and structural behavior of the Itinera system. These models help explain how different users and system modules interact with each other to perform major operations such as AI trip generation, marketplace purchasing, partner listing management, and admin monitoring.

4.4.1 Sequence Diagram

Sequence diagrams are interaction diagrams that show how different components of the Itinera platform communicate with each other in a specific workflow. They represent the order of messages exchanged between actors, user interfaces, backend modules, external APIs, and the database over time.

4.4.1.1 Sequence Diagram for User Core Process

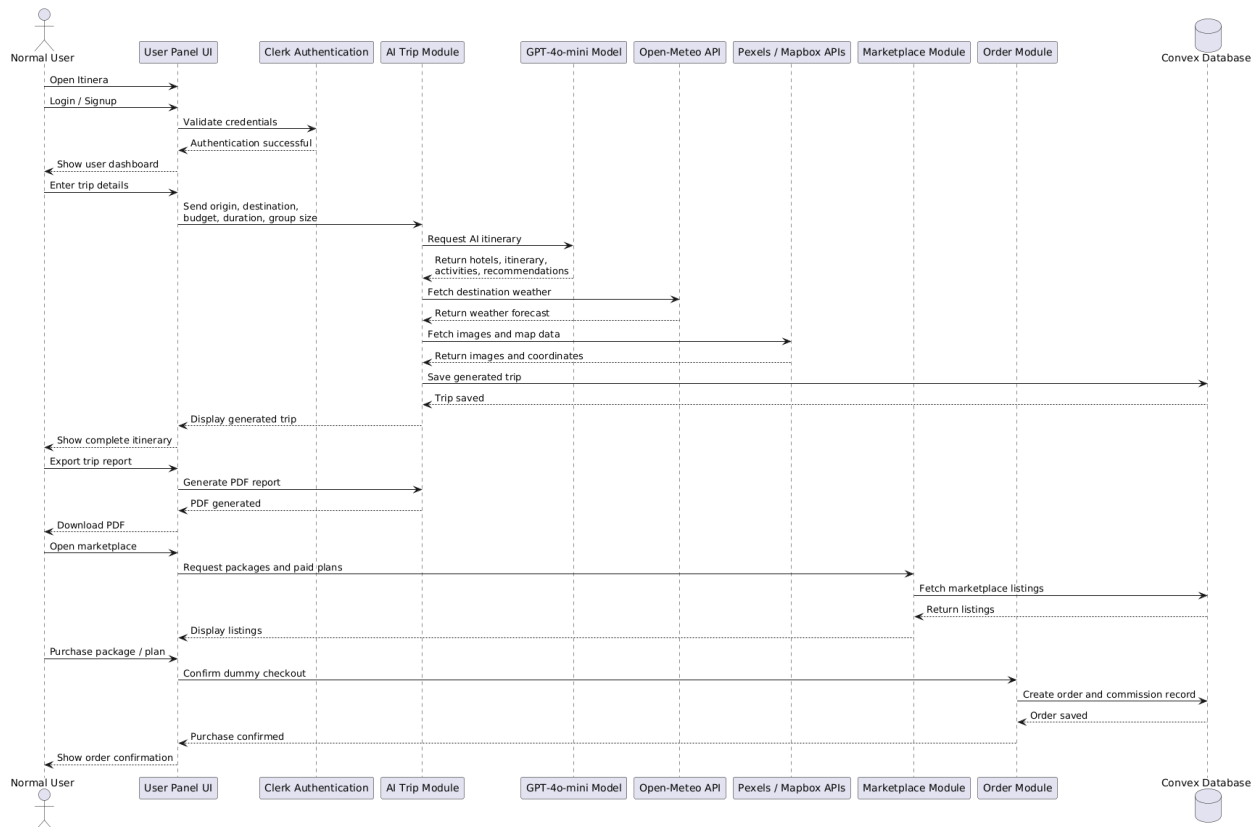


Figure 4.6: Sequence Diagram for User Core Process

Figure 4.6 shows the complete user workflow in Itinera, including authentication, AI trip generation, travel support, PDF export, marketplace browsing, and dummy checkout.

Key Interaction Phases and Data Flow:

- **Authentication Phase:** The workflow begins when the user accesses the platform and submits login or signup credentials through the User Panel UI. The interface forwards this request to the Clerk Authentication module, which independently validates the credentials. Upon successful verification, an authentication success message is returned, and the UI redirects the user to their secure personal dashboard.
- **AI Trip Generation Phase:** Once authenticated, the user initiates the planning process by entering specific trip parameters (origin, destination, budget, duration, and group size). The UI transmits this payload to the AI Trip Module. This module acts as the central orchestrator, executing a series of parallel and sequential external API calls:
 - It sends a prompt to the GPT-4o-mini Model to generate the core itinerary, hotel suggestions, and activities.
 - It queries the Open-Meteo API to fetch live destination weather forecasts.
 - It requests spatial and visual data from the Pexels / Mapbox APIs to retrieve map coordinates and high-quality location images. Once all external data is returned and aggregated, the AI Trip Module persists the complete trip record into the Convex Database. After the database confirms the save, the UI renders the complete, structured itinerary for the user.
- **Export and Utility Phase:** While viewing the generated itinerary, the user can trigger the "Export trip report" action. The AI Trip Module processes the saved itinerary data to generate a formatted PDF document. Once the PDF generation is complete, the file is returned to the UI, prompting the user to download the final report.
- **Marketplace and Checkout Phase:** The user transitions to the e-commerce side of the application by opening the marketplace via the UI. The interface requests available travel packages and paid plans from the Marketplace Module, which fetches active listings directly from the Convex Database and displays them. When the user selects a

listing and proceeds to "Purchase package / plan," the request is routed to the Order Module. The Order Module processes the dummy checkout and instructs the Convex Database to create a new order and log the corresponding commission record. Finally, the UI presents a purchase confirmation to the user, successfully concluding the sequence.

4.4.1.2 Sequence Diagram for Partner Core Process

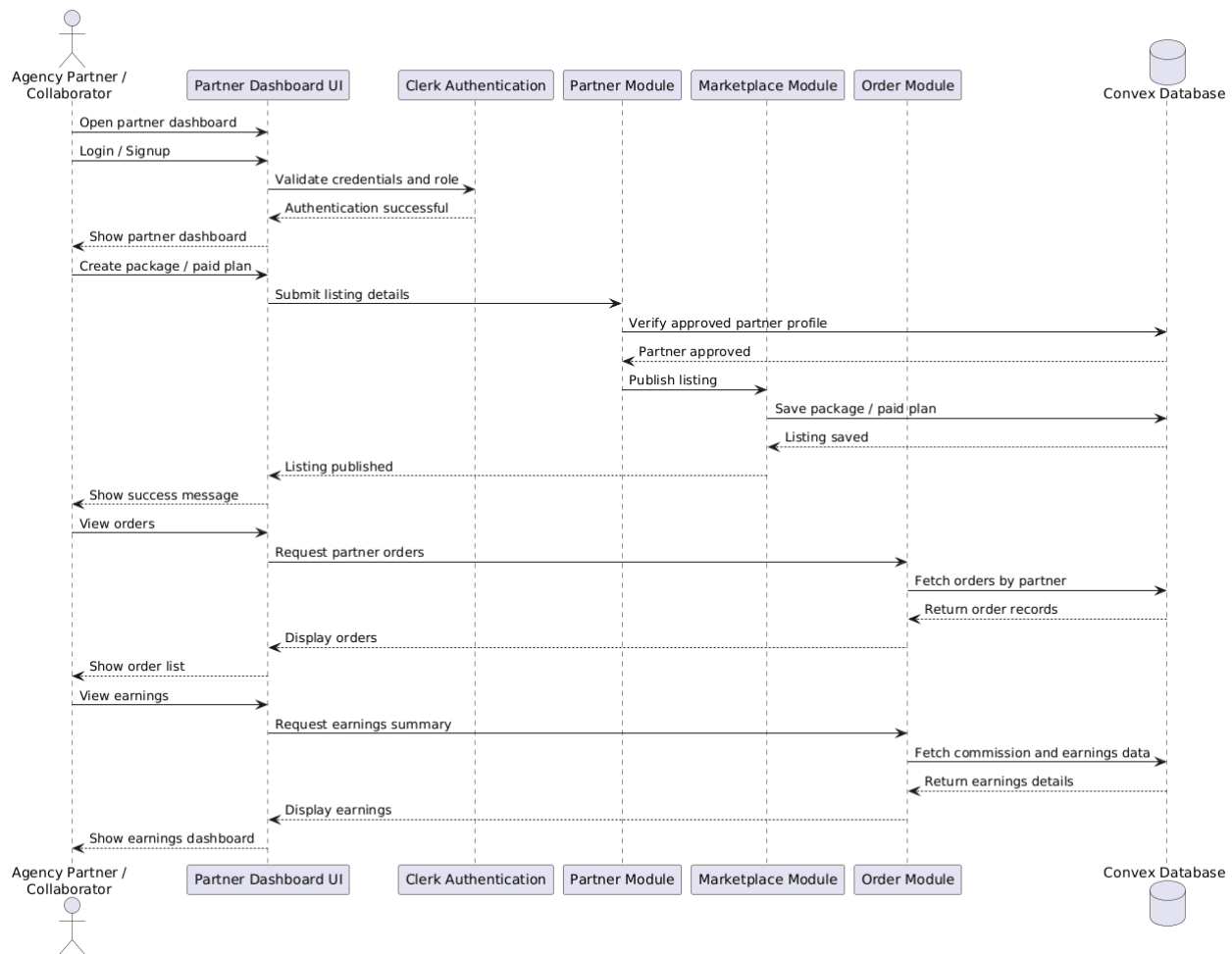


Figure 4.7: Sequence Diagram for Partner Core Process

Figure 4.7 illustrates how agency partners and collaborators manage marketplace listings, view customer orders, and track earnings in Itinera.

Key Interaction Phases and System Workflow:

- **Secure Authentication and Authorization Phase:** The sequence initiates when an Agency Partner or Collaborator accesses the platform and attempts to log in via the Partner Dashboard UI. The UI transmits the provided credentials to the Clerk Authentication module. Unlike normal user login, this step inherently validates both the credentials and the specific partner role. Once authentication is successful, the system grants access, and the UI renders the secure partner dashboard.
- **Listing Verification and Publication Phase:** When the partner decides to publish a new travel package or paid travel plan, they submit the listing details through the UI. The request is initially routed to the Partner Module. Crucially, before any data is published, the Partner Module communicates with the Convex Database to verify that the partner's profile holds an "approved" status. Upon database confirmation of this status, the Partner Module delegates the publishing task to the Marketplace Module. The Marketplace Module then commits the new listing to the Convex Database. Once successfully saved, a cascading confirmation returns to the UI, displaying a success message to the partner.
- **Order Tracking Phase:** To fulfill customer purchases, the partner navigates to the orders section of their dashboard. The Partner Dashboard UI sends a request to the Order Module. The Order Module queries the Convex Database to fetch all order records specifically associated with that partner's unique ID. The database returns these records, which are then formatted and displayed as an order list on the partner's UI.
- **Financial Tracking and Earnings Phase:** The final workflow illustrates the financial transparency of the platform. When the partner requests to view their earnings, the UI prompts the Order Module to compile an earnings summary. The Order Module retrieves the relevant commission and net earnings data from the ledger within the Convex Database. This financial data is returned to the frontend and rendered visually on the partner's earnings dashboard, completing the sequence of core partner operations.

4.4.1.3 Sequence Diagram for Admin Core Process

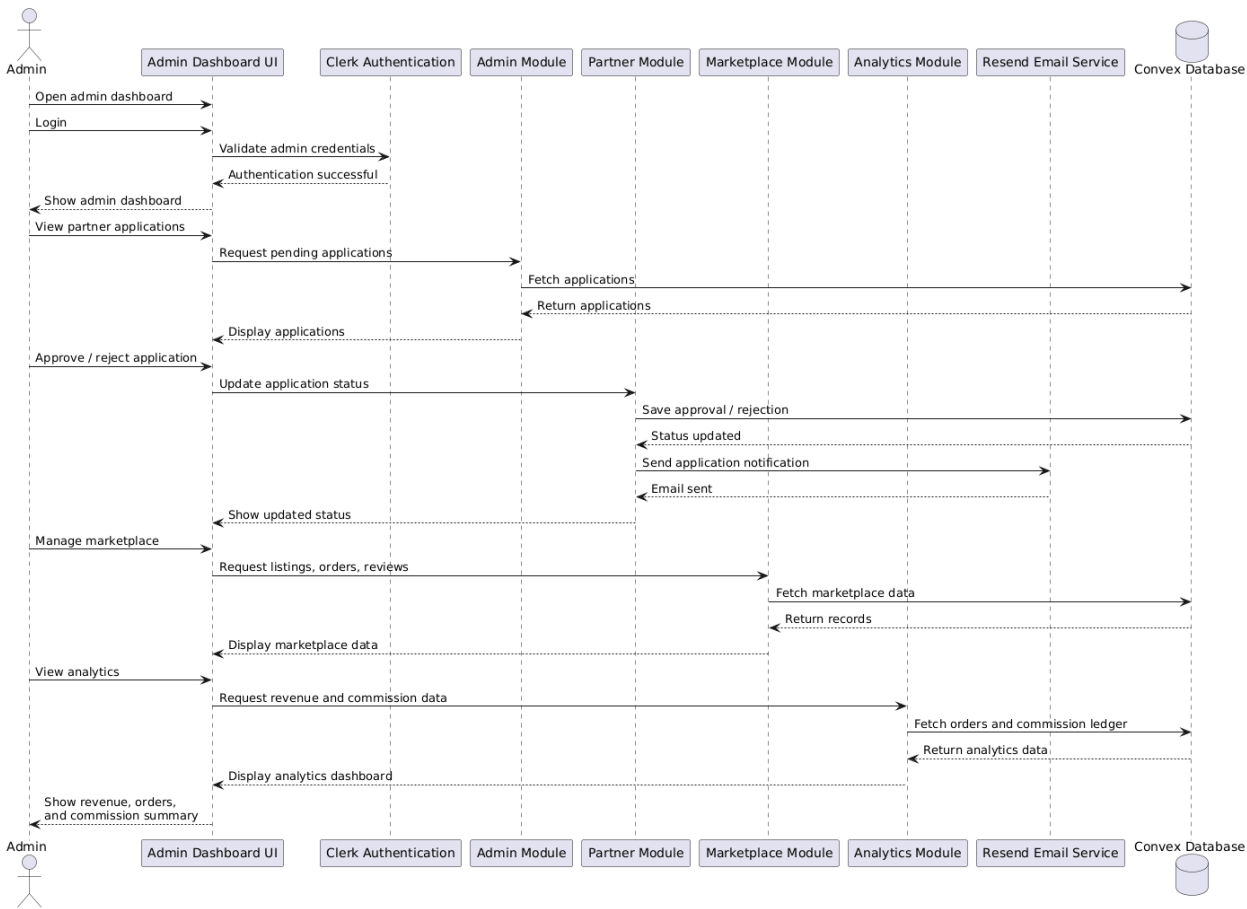


Figure 4.8: Sequence Diagram for Admin Core Process

Figure 4.8 shows the admin’s main responsibilities in Itinera, including partner approval, marketplace monitoring, order and review management, and revenue analytics.

Key Interaction Phases and System Workflow:

- **Secure Authentication and Session Initialization:**

The workflow begins with the administrator navigating to the platform and submitting login credentials through the Admin Dashboard UI. This request is securely routed to the Clerk Authentication module. The system strictly validates both the credentials and the user's

elevated administrative role. Upon successful verification, an authentication token is issued, and the UI successfully renders the secure administrative control panel.

- **Partner Application Review and Notification Workflow:** To manage the supply side of the platform, the admin requests to view pending partner applications. The UI sends a request to the Admin Module, which in turn fetches the pending records from the Convex Database. The data is returned and displayed on the dashboard. When the admin makes a decision (e.g., "Approve / reject application"), the UI forwards this command to the Partner Module. The module saves the updated status in the database. Furthermore, this action triggers an asynchronous external event: the module commands the Resend Email Service to dispatch an automated notification to the applicant regarding their approval status. Once the email is sent, the UI updates to reflect the resolved application status.
- **Marketplace Moderation and Monitoring:** To maintain the quality and integrity of the platform's e-commerce operations, the admin accesses the marketplace management interface. The Admin Dashboard UI sends a request to the Marketplace Module. This module executes a comprehensive read operation against the Convex Database to fetch all active listings (packages and paid plans), completed orders, and user reviews. The database returns these aggregated records, which are then rendered on the admin's screen for monitoring and moderation.
- **Financial Analytics and Revenue Tracking:** In the final illustrated workflow, the admin requests to view platform analytics to evaluate business performance. The UI triggers the Analytics Module, which requests raw revenue and commission data. The module queries the Convex Database specifically targeting the order records and the commission ledger. Once the financial data is fetched and aggregated, it is returned to the frontend. The UI processes this data to display a comprehensive analytics dashboard, summarizing total revenue, order volume, and platform commission.

4.4.2 Class Diagram

A class diagram is a static UML diagram that represents the structure of the Itinera system by showing its main classes, attributes, methods, and relationships. It serves as the foundational

blueprint for the platform's object-oriented architecture, helping to explain how different core entities such as users, trips, marketplace listings, orders, reviews, partner applications, and commission records are interconnected within the system.

In Itinera, the class diagram includes key classes such as User, Normal User, Admin, Agency Partner, Collaborator, Trip, Travel Package, Paid Travel Plan, Order, Review, Partner Application, and Commission Ledger. The diagram prominently features inheritance relationships to handle role-based access control efficiently; different specialized user roles are directly derived from a central, generalized User base class. This hierarchical approach ensures that shared attributes (like credentials and profile data) are centralized, while role-specific methods such as an admin approving applications, a user generating a trip, or a partner publishing a listing are strictly encapsulated within their respective subclasses.

Furthermore, the diagram maps out the critical operational relationships and multiplicities across the platform. It visually represents the behavioral flow of the system: it shows how users generate customized trips, how authorized partners publish marketplace listings, and how users interact with the e-commerce module to purchase packages or plans. It also details the transactional data flow by demonstrating how a completed Order triggers the creation of a Review object and automatically records financial splits through the Commission Ledger.

Overall, the class diagram provides a clear structural view of the Itinera platform and supports a better understanding of how data and logic are organized for AI trip planning, marketplace management, partner operations, and admin control. By explicitly defining these entities, their internal states, and their associations, the diagram bridges the gap between the conceptual software design and the actual schema implementation within the platform's convex database.

4.4.2.1 Class Diagram

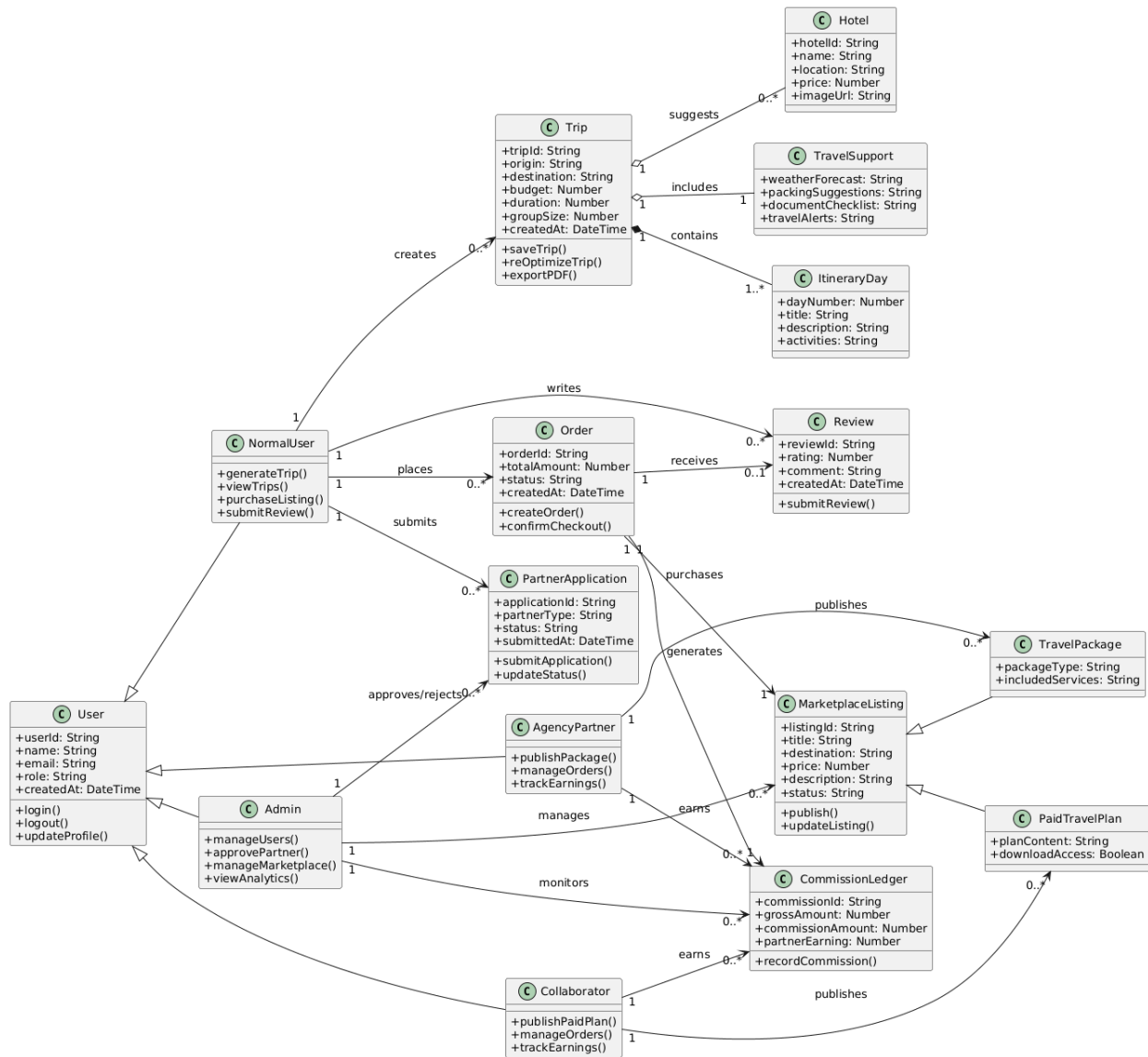


Figure 4.9: Class Diagram of Itinera System

Figure 4.9 represents the core classes of the Itinera platform and their relationships. It shows how users generate trips, partners publish marketplace listings, users place orders and reviews, and the system records commission through the marketplace model.

Key Architectural Modules and Class Relationships:

- User Hierarchy and Role-Based Access (Inheritance):** At the core of the system's identity management is the base User class, which encapsulates common attributes (userId, name, email, role) and authentication methods (login(), logout()). The system

utilizes generalization (inheritance) to derive four distinct user roles from this base class:

- **NormalUser:** Can trigger generateTrip(), purchaseListing(), and submitReview().
- **Admin:** Inherits core properties but includes elevated methods like manageUsers(), approvePartner(), and viewAnalytics().
- **AgencyPartner & Collaborator:** Specialized roles capable of publishing packages, managing orders, and tracking earnings.
- **Trip Generation Module (Composition):** The Trip class is a central entity created by the NormalUser. It holds parameters like budget, duration, and destination. This class exhibits strong *composition* relationships (indicated by solid diamonds) with structural sub-components:
 - **ItineraryDay:** A trip contains one or more days (1..*), detailing specific activities.
 - **TravelSupport:** A trip strictly includes one (1..1) travel support object holding weatherForecast and packingSuggestions.
 - Additionally, the Trip class has a standard association with the Hotel class, as it *suggests* zero or more (0..*) hotels based on the generated itinerary.
- **Marketplace and E-Commerce Module (Polymorphism):** The marketplace utilizes an abstract base class called MarketplaceListing that defines shared attributes (price, title, status). Using inheritance, this branches into two specific listing types:
 - **TravelPackage:** Published by AgencyPartners, containing includedServices.
 - **PaidTravelPlan:** Published by Collaborators, representing downloadable content (downloadAccess).
- **Transaction and Financial Flow (Associations):** When a NormalUser *places* an Order, it triggers a cascade of transactional relationships:
 - The Order class is directly linked to the MarketplaceListing it *purchases*.

- Once an order is confirmed, it *generates* exactly one (1..1) CommissionLedger record. This ledger accurately splits the totalAmount into commissionAmount (for the platform) and partnerEarning.
- AgencyPartners and Collaborators are linked to this CommissionLedger to *earn* their respective revenue.
- **Feedback and Application Workflows:**
 - **Review Class:** A NormalUser can *write* a review, which is explicitly tied to an Order (an order *receives* a review), ensuring only verified buyers can leave feedback.
 - **PartnerApplication Class:** Normal users seeking to become partners *submit* applications. The Admin class is associated with this entity to *approve/reject* the status, which dynamically upgrades the user's role upon approval.

Chapter 5

IMPLEMENTATION

5 Implementation

In the implementation phase of Itinera, the core features of the platform were developed through a set of structured algorithms and integrated modules. These algorithms support AI trip generation, trip re-optimization, weather-based travel assistance, PDF report export, marketplace operations, dummy checkout, reviews, partner approval, and commission tracking. The system is built using Next.js 15, React, TypeScript [6], Tailwind [8] CSS, Convex, Clerk Authentication, OpenAI SDK, Open-Meteo API, Pexels API, Mapbox, Arcjet [10], and Resend. The following section follows the same implementation structure as the provided sample.

5.1 Algorithm

The following subsections describe the major algorithms implemented in Itinera. Each algorithm includes its function name, input, output, description, and pseudocode.

5.1.1 User Authentication/Login

Function: `authenticateUser(userCredentials)`

Input:

`userCredentials` email, password, role

Output:

Authenticated user session or error message

Description:

This algorithm handles secure login and signup for Normal Users, Admins, Agency Partners, and Collaborators using Clerk Authentication. After successful authentication, the system checks the user role and redirects the user to the appropriate dashboard.

Pseudocode:

- Receive email and password from the user.
- Send credentials to Clerk Authentication.
- If credentials are invalid, display login error.
- If credentials are valid, create user session.

- Fetch user role from the database.
- Redirect user to the correct dashboard based on role.
- Allow access only to authorized modules.

5.1.2 AI Trip Generation

Function: generateAITrip(tripDetails)

Input:

tripDetails origin, destination, budget, duration, groupSize

Output:

AI-generated trip itinerary or error message

Description:

This algorithm generates a personalized travel itinerary using GPT-4o-mini. The user provides basic trip details, and the AI model returns hotels, day-wise itinerary, activities, travel recommendations, locations, and coordinates.

Pseudocode:

- Receive origin, destination, budget, duration, and group size.
- Validate all required trip fields.
- If any field is missing, display validation error.
- Send trip details to the AI model.
- Generate hotel suggestions, itinerary, activities, and recommendations.
- Fetch images using Pexels API.
- Fetch location coordinates using Mapbox [7].
- Save generated trip in convex database.
- Display complete trip plan to the user.

5.1.3 AI Trip Re-Optimization

Function: reOptimizeTrip (tripId, instruction)

Input:

tripId, instruction

Output:

Updated trip plan or error message

Description:

This algorithm allows users to improve an existing trip using AI instructions. Users can request changes such as making the trip cheaper, reducing walking, adding rainy-day alternatives, making it family friendly, or optimizing it for food experiences.

Pseudocode:

- Receive selected trip ID and user instruction.
- Fetch existing trip from convex database.
- Validate that the trip exists.
- Send current itinerary and instruction to AI model.
- Generate updated itinerary according to instruction.
- Save updated trip plan in the database.
- Display re-optimized trip to the user.

5.1.4 Weather and Travel Support

Function: getTravelSupport(destination)

Input:

destination

Output:

Weather forecast, alerts, packing suggestions, and document checklist

Description:

This algorithm provides live destination weather and travel support using the Open-Meteo API. Based on weather conditions, the system generates packing suggestions, document checklist, and travel alerts.

Pseudocode:

- Receive destination from generated trip.
- Fetch destination weather using Open-Meteo API.
- Analyze temperature, rainfall, and weather conditions.
- Generate weather-based travel alerts.
- Create smart packing suggestions.
- Generate travel document checklist.
- Display travel support details in trip view.

5.1.5 PDF Travel Report Export

Function: `exportTripReport(tripId)`

Input:

`tripId`

Output:

Printable PDF travel report or error message

Description:

This algorithm creates a printable PDF report for a generated trip. The report includes trip summary, hotels, itinerary, activities, travel essentials, packing suggestions, and document checklist.

Pseudocode:

- Receive selected trip ID.
- Fetch trip details from Convex database.
- Validate that trip data exists.
- Format trip summary, hotels, itinerary, and activities.
- Add weather support, packing list, and checklist.
- Generate PDF report.
- Allow user to download or print the report.

5.1.6 Publish Marketplace Listing

Function: publishListing(partnerId, listingData)

Input:

partnerId, listingData

Output:

Published package/plan or error message

Description:

This algorithm allows approved Agency Partners to publish travel packages and approved Collaborators to publish paid travel plans. The listing is saved in the marketplace after validation.

Pseudocode:

- Receive partner ID and listing details.
- Verify that the partner is approved.
- Validate title, destination, price, description, and images.
- Identify listing type as travel package or paid travel plan.
- Save listing in convex database.
- Display listing in the marketplace.
- Show success message to partner or collaborator.

5.1.7 Dummy Checkout and Order Creation

Function: createOrder (userId, listingId)

Input:

userId, listingId

Output:

Order confirmation or error message

Description:

This algorithm manages the dummy checkout process. When a user purchases a marketplace

package or paid travel plan, the system creates an order record and records platform commission and partner earnings.

Pseudocode:

- Receive user ID and selected listing ID.
- Fetch listing details from the database.
- Validate that listing is available.
- Confirm dummy checkout.
- Create order record with user, listing, amount, and status.
- Calculate platform commission.
- Calculate partner earning.
- Save order and commission record in database.
- Display order confirmation to the user.

5.1.8 Reviews and Ratings

Function: submitReview(userId, orderId, reviewData)

Input:

userId, orderId, reviewData

Output:

Saved review or error message

Description:

This algorithm allows users to submit reviews and ratings for purchased marketplace listings.

The system verifies that the user has completed a valid order before accepting the review.

Pseudocode:

- Receive user ID, order ID, rating, and comment.
- Verify that the order belongs to the user.
- Check that the package or plan was purchased.

- Validate rating and review text.
- Save review in Convex database.
- Update listing review section.
- Display review on marketplace listing.

5.1.9 Partner Application Approval

Function: `managePartnerApplication(adminId, applicationId, action)`

Input:

`adminId, applicationId, action`

Output:

Updated application status or error message

Description:

This algorithm allows the admin to approve or reject applications submitted by users who want to become Agency Partners or Collaborators.

Pseudocode:

- Receive admin ID, application ID, and action.
- Verify that the user has admin role.
- Fetch partner application from database.
- Review submitted details.
- If approved, update application status.
- Assign role as Agency Partner or Collaborator.
- Create partner profile.
- Send approval email using Resend.
- If rejected, update status and notify applicant.

5.1.10 Commission and Earnings Tracking

Function: `calculateCommission(orderId)`

Input:

orderId

Output:

Commission record and partner earning

Description:

This algorithm calculates platform commission and partner earnings after a marketplace order is completed. The commission ledger helps the admin monitor revenue and provides partners with transparent earning records.

Pseudocode:

- Receive completed order ID.
- Fetch order amount and partner details.
- Apply platform commission percentage.
- Calculate partner earning after commission deduction.
- Save gross amount, commission amount, and partner earning.
- Update partner earnings dashboard.
- Display commission data in admin analytics.

5.2 External APIs

Table 5.1: External APIs

Library / API	Purpose
Next.js API Routes	Provides backend endpoints for AI requests, marketplace operations, orders, and email handling.
Convex	Handles backend functions and database operations for users, trips, packages, orders, reviews, and commissions.
Clerk Authentication	Provides secure login, signup, session management, and role-based access control.

OpenAI SDK / GPT-4o-mini	Used for AI trip generation and AI trip re-optimization.
Open-Meteo API	Provides live destination weather forecasts and weather-based travel support.
Pexels API	Fetches destination, hotel, and place images for travel plans.
Mapbox API	Provides location coordinates and map-related support.
Resend	Sends email notifications such as partner approval or order-related emails.
Arcjet	Provides security protection against suspicious requests and abuse.
ShadCN UI	Provides reusable UI components for dashboards, forms, and marketplace screens.
Tailwind CSS	Used for responsive and modern interface styling.

5.3 User Interface

The Itinera platform is designed with a user-centered approach to provide a simple, modern, and responsive experience. The interface is divided into role-based sections for Normal Users, Admins, Agency Partners, and Collaborators. Each role is provided with relevant features such as AI trip generation, trip management, marketplace browsing, order handling, partner publishing, and admin monitoring.

The user interface allows normal users to sign in, generate personalized trips, view AI-generated itineraries, access weather support, export PDF reports, browse marketplace listings, purchase travel packages, and submit reviews. Admins can manage users, approve partner applications, monitor marketplace activity, and view commission analytics. Partners and collaborators can publish listings, manage orders, and track earnings.

The complete interface is responsive and works across desktops, laptops, tablets, and mobile devices. A clean layout, structured forms, clear buttons, and dashboard-based navigation help users interact with the system easily.

5.3.1 Sign-in / Sign-up Screen

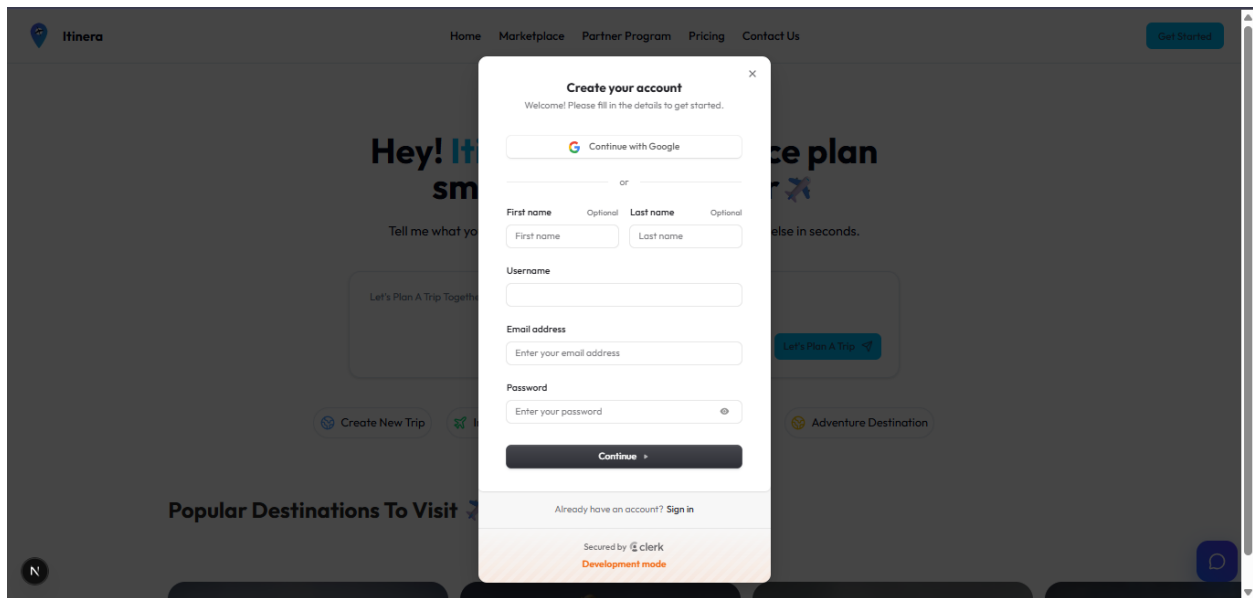


Figure 5.1: Sign-in / Sign-up Screen

This screen allows users to securely log in or create an account in Itinera using Clerk Authentication. After successful authentication, the system redirects users to their respective dashboard based on their role, such as Normal User, Admin, Agency Partner, or Collaborator.

Key Interface Components and Features:

- **Dual Authentication Methods:** The interface offers users the flexibility to choose between Single Sign-On (SSO) and traditional email registration. A prominent "Continue with Google" button sits at the top of the form, allowing users to bypass manual entry and authenticate instantly using their existing Google credentials.
- **Structured Form Input:** For users opting for manual registration, the form is cleanly organized with dedicated input fields. It collects basic profile information through optional "First name" and "Last name" fields placed side-by-side, followed by required

fields for "Username," "Email address," and a securely masked "Password" field equipped with a visibility toggle icon.

- **Clear Call to Action:** A distinct, high-contrast "Continue" button is situated at the bottom of the form to submit the credentials. Below this, a standard navigation link ("Already have an account? Sign in") allows existing users to easily switch from the registration flow to the login flow.
- **Security Integration:** At the footer of the modal, a "Secured by Clerk" badge is visibly displayed. This confirms the platform's integration with Clerk Authentication, a robust identity management provider that handles the backend security, encryption, and session management. *(Note: The "Development mode" indicator reflects the current staging environment of the application).*
- **Post-Authentication Routing:** Once the user successfully completes the sign-in or sign-up process, the system dynamically assesses the user's profile state. Based on their assigned role in the database, the authentication controller seamlessly redirects them to their respective workspace whether it is the standard User Dashboard, the Admin Control Panel, or the dedicated portals for Agency Partners and Collaborators.

5.3.2 AI Trip Generator Screen

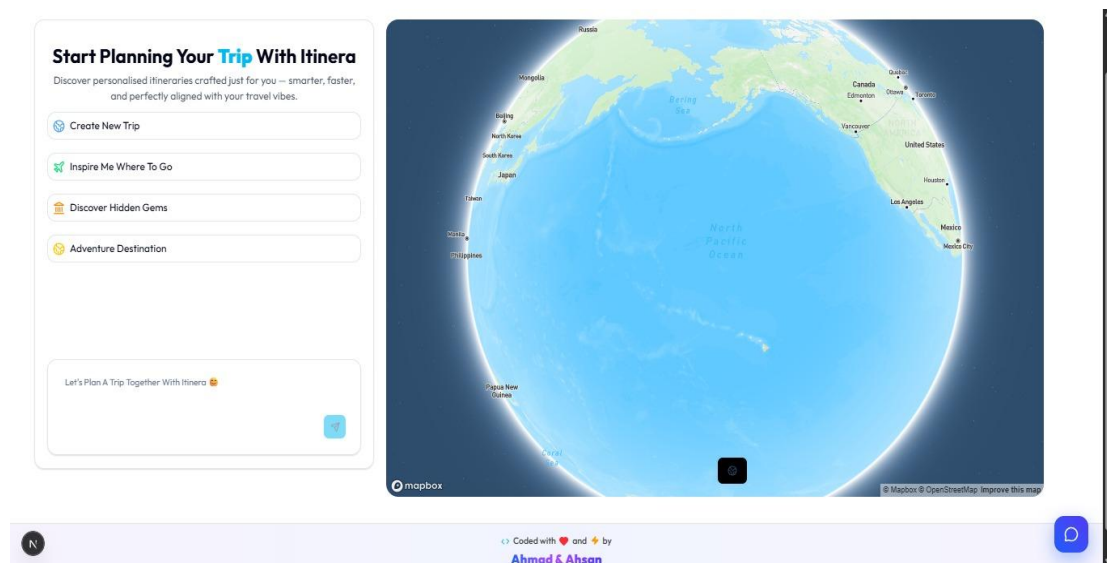


Figure 5.2: AI Trip Generator Screen

This screen allows users to enter travel details including origin, destination, budget, duration, and group size. After submitting the form, the system sends the input to the AI model and generates a personalized travel itinerary.

Key Interface Components and Features:

- **Dual-Mode Interactive Inputs:** The left panel provides a highly accessible, dual-input approach to trip planning. Users can utilize the quick-start prompt buttons (e.g., "Create New Trip," "Inspire Me Where To Go," "Discover Hidden Gems," and "Adventure Destination") for guided, structured planning. Alternatively, users can utilize the natural language text field at the bottom ("Let's Plan A Trip Together With Itinera") to specify detailed travel parameters such as origin, destination, budget, duration, and group size in a conversational format.
- **Immersive Geospatial Visualization:** The right side of the interface is dedicated to a dynamic, interactive 3D globe powered by Mapbox. This prominent geographical component enhances the user experience by providing immediate visual context for their travel aspirations, creating a strong connection between the planning phase and the physical destinations.
- **AI Processing Pipeline:** The design facilitates a seamless transition from user intent to system action. Upon submitting the travel details through the input field, the frontend captures the data payload and securely transmits it to the underlying AI model. The system then processes these specific constraints and preferences to autonomously compute and generate a comprehensive, personalized travel itinerary.
- **Modern UX Architecture:** The screen is built with clean typography, clear spacing, and intuitive visual hierarchy. Additionally, the footer features a floating support widget in the bottom right corner, ensuring users have continuous access to assistance without leaving the active planning environment.

5.3.3 Generated Trip Details Screen

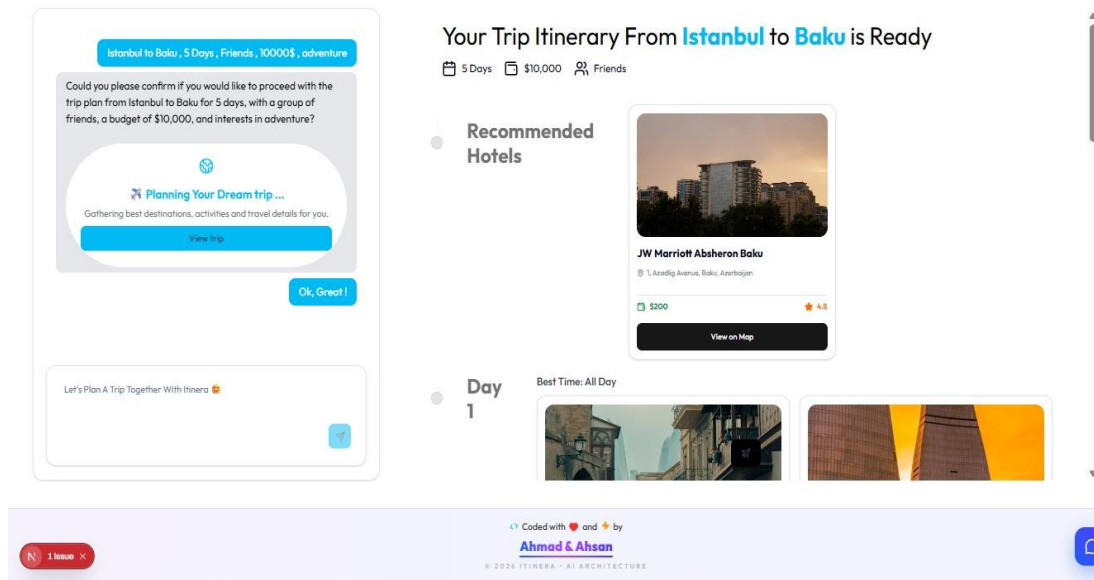


Figure 5.3: Generated Trip Details Screen

This screen displays the complete AI-generated trip plan, including hotel suggestions, day-wise itinerary, activities, travel recommendations, weather forecast, packing suggestions, and document checklist. Users can also re-optimize the trip or export it as a PDF report.

Key Interface Components and Features:

- **Conversational Context and Refinement Panel (Left):** The left side of the screen retains a chat-based interface that displays the historical context of the user's request (e.g., "Istanbul to Baku, 5 Days, Friends, 10000\$, adventure"). This pane not only confirms the system's understanding of the parameters but also includes an active input field at the bottom. This allows users to iteratively converse with the AI to dynamically **re-optimize** the trip such as requesting cheaper alternatives, changing the pace of the trip, or adding specific points of interest without starting over.
- **Structured Itinerary Presentation (Right):** The primary focal point is the scrollable right pane, which houses the complete, personalized travel plan. It begins with a clear, dynamic header confirming the trip's origin, destination, and key metadata (duration, budget, and group type).

- **Chronological and Categorized Data Delivery:** The AI-generated data is rendered into highly readable, card-based UI components organized along a vertical timeline:
 - **Recommended Hotels:** Displays curated accommodation options tailored to the user's budget. Each card features high-quality imagery (fetched via external APIs), specific pricing, user ratings, and a direct "View on Map" integration for geospatial context.
 - **Day-wise Itinerary:** The timeline breaks the trip down chronologically (e.g., "Day 1"), presenting scheduled activities, recommended visiting times ("Best Time: All Day"), and visual previews of the destinations.
- **Comprehensive Travel Support & Export Capabilities:** As users scroll through this pane, the system further presents the integrated travel support modules. This includes contextual weather forecasts, smart packing suggestions, and necessary travel document checklists based on the specific destination. Additionally, the interface supports functionality to export this entire structured view into a formatted, printable **PDF report** for offline accessibility.

5.3.4 Marketplace Screen

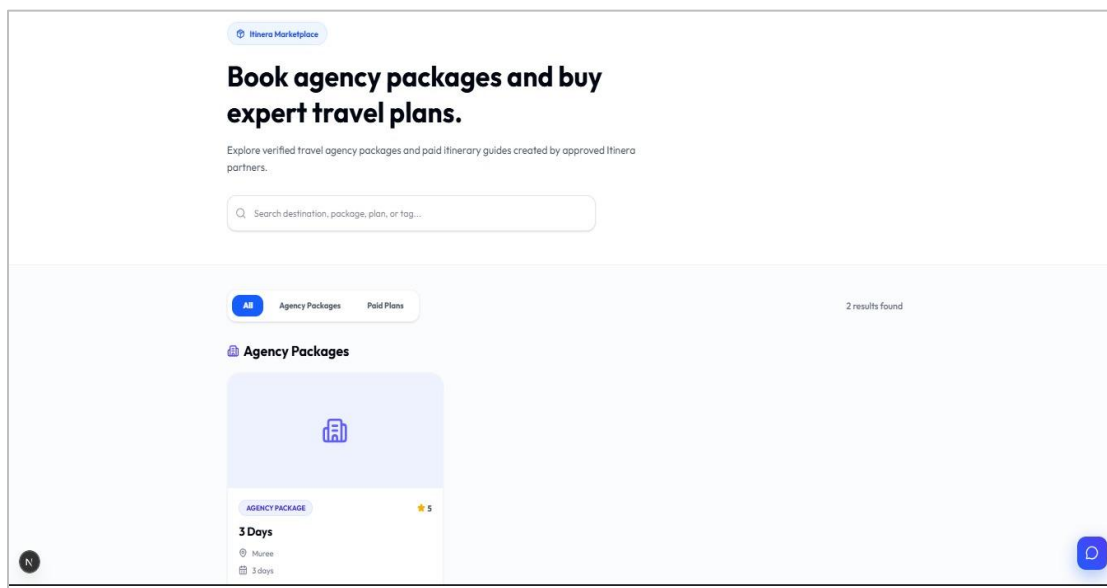


Figure 5.4: Marketplace Screen

This screen shows travel packages and paid travel plans published by agency partners and collaborators. Users can browse listings, view details, purchase packages or plans through dummy checkout, and later submit reviews and ratings.

Key Interface Components and Features:

- **Hero Section and Global Search:** The top of the screen features a clear, bold value proposition: "Book agency packages and buy expert travel plans," immediately informing the user of the page's purpose. Below this, a prominent, full-width search bar allows users to perform targeted queries based on specific destinations, package names, travel plans, or thematic tags, ensuring quick access to relevant content.
- **Dynamic Filtering and Categorization:** To manage different types of marketplace listings, the interface employs a segmented control mechanism (tabbed navigation) offering filters for "All," "Agency Packages," and "Paid Plans." This allows users to seamlessly switch between full-service travel agency offerings and expert itinerary guides published by travel bloggers or collaborators. A dynamic results counter (e.g., "2 results found") updates in real-time based on the active filters.
- **Structured Listing Cards:** The marketplace inventory is displayed using a responsive grid of card components. Each listing card is designed for quick scanning, presenting essential metadata at a glance. Features include a visual thumbnail, an entity badge (e.g., "AGENCY PACKAGE"), an aggregate star rating, a bold package title (e.g., "3 Days"), and geographical/temporal details (e.g., location "Muree" and duration).
- **Integrated Transaction and Feedback Loop:** While not fully visible in the high-level grid view, clicking on any of these listing cards routes the user to a detailed product page. From there, the UI supports the platform's core commercial workflows: users can proceed through a **dummy checkout** process to simulate purchasing a package or plan. Furthermore, post-purchase, the interface provides mechanisms for users to submit verified **reviews and ratings**, which dynamically update the star ratings displayed on these very cards.

5.3.5 Admin Dashboard Screen

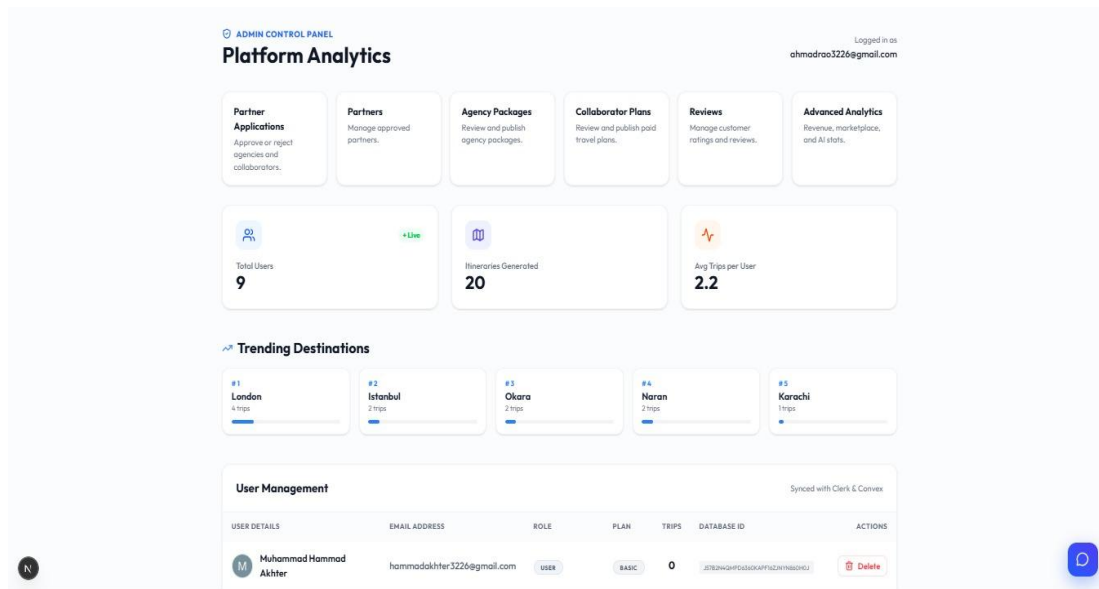


Figure 5.5: Admin Dashboard Screen

This screen represents the management side of Itinera. Admins can approve partner applications, manage users, monitor marketplace activity, and view analytics, while agency partners and collaborators can manage listings, orders, and earnings.

Key Interface Components and Features:

- **Executive Navigation and Module Access:** The top section of the dashboard features a grid of high-level action cards that act as gateways to specific management modules. From here, administrators can navigate to "Partner Applications" to approve or reject prospective agencies and collaborators, manage existing "Partners," review published "Agency Packages" and "Collaborator Plans," moderate customer "Reviews," and access "Advanced Analytics" for deep-dive revenue and AI utilization statistics.
- **Real-Time Key Performance Indicators (KPIs):** Below the navigation grid, the system highlights critical platform health metrics through bold stat cards. This includes tracking "Total Users" (complete with a dynamic "+ Live" indicator to signify real-time

data sync), total "Itineraries Generated" by the AI engine, and the "Avg Trips per User." These metrics give admins instant insight into platform growth and user engagement.

- **Analytical Insights (Trending Destinations):** The interface includes a visually engaging "Trending Destinations" section. This ranks the most frequently searched or generated travel locations (e.g., London, Istanbul, Okara, Naran, Karachi) using bar charts. This real-time analytics feature helps platform owners understand market demand and user travel preferences.
- **Integrated User Management Matrix:** The lower half of the screen houses a robust data table dedicated to "User Management," which explicitly notes its synchronization with backend services ("Synced with Clerk & Convex"). The table provides granular visibility into user records, displaying their Name, Email Address, assigned Role (e.g., USER, ADMIN), subscription/account Plan, total Trips, and a unique Database ID. It also includes actionable controls, such as a "Delete" function, enabling administrators to enforce platform security and perform necessary CRUD (Create, Read, Update, Delete) operations directly from the UI.
- **Extensible Partner Infrastructure:** While this specific view is tailored for the platform Administrator, the underlying dashboard architecture is similarly utilized to provide specialized, role-based views for Agency Partners and Collaborators. In those parallel dashboards, partners are equipped with targeted tools to publish listings, process order fulfillment, and track their commission-based earnings securely.

Chapter 6

TESTING AND EVALUATION

6 Testing and Evaluation

Testing plays an important role in ensuring that the Itinera platform works correctly, meets user requirements, and provides a reliable travel planning experience. Testing was performed to identify bugs, validate workflows, and confirm smooth integration between AI trip generation, authentication, marketplace, weather support, PDF export, and admin/partner dashboards. This section follows the testing structure provided in the sample report.

6.1 Manual Testing

Manual testing was conducted to evaluate the interface, navigation flow, and main features of the Itinera platform. Testers interacted directly with the system to verify that each feature worked according to the defined requirements.

During manual testing, login and signup were tested for Normal Users, Admins, Agency Partners, and Collaborators. The AI trip generation workflow was tested by entering origin, destination, budget, duration, and group size. The generated itinerary, hotel suggestions, weather forecast, packing suggestions, and PDF report export were also verified.

Marketplace testing included browsing travel packages, viewing paid travel plans, completing dummy checkout, creating orders, and submitting reviews. Partner-side testing included publishing packages, managing listings, viewing orders, and tracking earnings. Admin-side testing included approving partner applications, managing users, monitoring marketplace activity, and viewing revenue analytics.

6.1.1 System Testing

System testing was performed after integrating all major modules to verify the overall behavior of the Itinera platform.

Table 6.1: System Testing Table

Test ID	Feature Tested	Description	Expected Result	Actual Result	Status
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ST-01	User Authentication	Test login/signup for all roles	Successful login and role-based redirection	As expected,	Pass
ST-02	AI Trip Generation	Generate trip using user inputs	AI itinerary is generated	As expected,	Pass
ST-03	Weather Support	Fetch destination weather	Weather forecast is displayed	As expected,	Pass
ST-04	PDF Export	Export generated trip report	PDF report is generated	As expected,	Pass
ST-05	Marketplace Purchase	Test dummy checkout and order creation	Order and commission record created	As expected,	Pass

Table 6.1 summarizes the system testing results. Each test case validates a major feature of Itinera, and all core modules functioned as expected.

6.1.2 Unit Testing

Unit testing was performed on individual modules to ensure that each component worked correctly before integration.

Table 6.2: Unit Function Table

Module	Function	Expected Output	Actual Output	Status
--------	----------	-----------------	---------------	--------

Authentication	authenticateUser()	Valid user session created	Session created successfully	Pass
AI Trip Module	generateAITrip()	Trip itinerary generated	Generated successfully	Pass
Re-Optimizer Module	reOptimizeTrip()	Updated trip plan returned	Updated successfully	Pass
Marketplace Module	publishListing()	Listing saved in marketplace	Saved successfully	Pass
Order Module	createOrder()	Order and commission created	Created successfully	Pass
Review Module	submitReview()	Review saved	Saved successfully	Pass

Table 6.2 shows that individual functions performed according to their expected outputs before being integrated into the complete system.

6.1.3 Functional Testing

Functional testing ensured that all major use cases of Itinera worked correctly for users, admins, agencies, and collaborators.

Table 6.3: Functional Testing Table

Test Case ID	Use Case	Input	Expected Output	Actual Output	Status
TC-01	Register / Login	Valid credentials	Redirect to dashboard	Successful login	Pass
TC-02	Generate AI Trip	Trip details	Personalized itinerary displayed	As expected	Pass
TC-03	Re-Optimize Trip	AI instruction	Updated itinerary generated	As expected	Pass
TC-04	Browse Marketplace	Open marketplace	Packages and plans displayed	As expected	Pass
TC-05	Purchase Plan	Select listing and checkout	Order confirmed	As expected	Pass
TC-06	Submit Review	Rating and comment	Review saved	As expected	Pass

Table 6.3 presents functional test cases based on the main user interactions of the Itinera platform.

6.1.4 Integration Testing

Integration testing was performed to ensure smooth communication between interconnected modules.

Tested Integrations:

- Clerk Authentication ↔ Role-Based Dashboards
- AI Trip Module ↔ GPT-4o-mini Model
- Trip Module ↔ Open-Meteo [11], Pexels, and Mapbox APIs
- Marketplace Module ↔ Order and Commission Module
- Partner Dashboard ↔ Marketplace Listings
- Admin Dashboard ↔ Partner Approval and Analytics

Table 6.4: Integration Testing Table

Integration ID	Modules Integrated	Expected Result	Actual Result	Status
IN-01	Authentication and Dashboards	User redirected by role	As expected	Pass
IN-02	AI Model and Trip Module	Trip generated successfully	As expected	Pass
IN-03	Weather API and Trip Details	Weather displayed correctly	As expected	Pass
IN-04	Marketplace and Order Module	Order created after checkout	As expected	Pass
IN-05	Order and Commission Module	Commission recorded correctly	As expected	Pass

Table 6.4 shows that the integrated modules communicated correctly and maintained consistent data flow.

6.2 Tools and Technologies

Table 6.5: Tools and Technologies Table

Tool / Framework	Category	Purpose / Description
Next.js 15	Full Stack Framework	Handles frontend rendering and API routes
React	Frontend Library	Builds responsive user interfaces
TypeScript	Programming Language	Provides type-safe development
Tailwind CSS	UI Styling	Creates responsive and modern layouts
ShadCN UI	UI Components	Provides reusable interface components
Convex	Backend and Database	Handles backend functions and data storage
Clerk Authentication	Authentication	Manages login, signup, and role-based access
OpenAI SDK / GPT-4o-mini	AI Integration	Generates and re-optimizes travel itineraries
Open-Meteo API	Weather API	Provides destination weather forecasts
Pexels API	Image API	Fetches travel and hotel images
Mapbox API	Map API	Provides coordinates and map support
Postman	API Testing Tool	Tests backend endpoints
Git & GitHub	Version Control	Supports code management and collaboration

Table 6.5 lists the major tools and technologies used during development and testing of the Itinera platform.

The testing and evaluation phase confirmed that the Itinera platform is functional, stable, and ready for deployment. Manual testing, unit testing, functional testing, and integration testing were performed to validate all major features, including authentication, AI trip generation, trip re-optimization, weather support, PDF export, marketplace browsing, dummy checkout, reviews, partner dashboards, and admin analytics.

Minor interface and validation issues identified during testing were fixed to improve usability. Overall, the testing phase verified that Itinera meets its project objectives by providing a secure, intelligent, and user-friendly AI travel planning and marketplace platform.

Chapter 7

CONCLUSION AND FUTURE WORK

Conclusion

The development of the Itinera platform successfully achieved its main objective to design and implement a modern, AI-powered travel planning and marketplace system. The platform simplifies the travel planning process by allowing users to generate personalized itineraries based on origin, destination, budget, duration, and group size. It combines AI trip generation, weather-based travel support, smart packing suggestions, travel document checklists, PDF report export, and marketplace functionality into one integrated web application.

Throughout the project, strong focus was placed on usability, scalability, and modular system design. The platform was divided into different modules such as authentication, AI trip generation, trip management, marketplace, orders, reviews, partner dashboards, admin management, and commission tracking. This modular structure made the system easier to develop, test, maintain, and extend in the future.

7.1 Future Work

Although the current version of Itinera fulfills its core objectives, several enhancements can further improve its functionality and real-world usability.

- **Real Payment Gateway Integration:**
Integrating secure payment gateways such as Stripe, PayPal, or local payment methods would allow users to purchase travel packages and paid plans through real transactions instead of dummy checkout.
- **Mobile Application Development:**
Dedicated Android and iOS applications can be developed to make trip planning more accessible and convenient for users on mobile devices.
- **Flight and Hotel Booking Integration:**
Future versions can integrate real flight and hotel booking APIs so users can directly book travel services from within the platform.
- **Advanced AI Recommendations:**
The AI model can be improved to provide more personalized suggestions based on user interests, previous trips, travel style, food preferences, and seasonal trends.
- **Real-Time Chat System:**

A chat feature can be added to allow users to communicate directly with agency partners or collaborators before purchasing a package or plan.

➤ Enhanced Analytics Dashboard:

More detailed analytics can be added for admins and partners, including sales trends, popular destinations, user behavior, and revenue forecasting.

Chapter 8

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